A Semantic Mutation Metric for Metamorphic-Relation Adequacy in Scientific Computing Programs

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Metamorphic testing mitigates the test-oracle problem by checking relations among executions, but classical mutation score is defined over syntactic program edits and does not say whether a metamorphic-relation set observes declared domain-semantic effects. This paper proposes Semantic Mutation Score (SMS), a backward-compatible adequacy metric for metamorphic-relation sets over certified semantic-effect strata. SMS keeps the denominator discipline of classical mutation testing while making mutant generation, equivalence, and killing relative to an explicit semantic certificate. We instantiate five semantic operator families on 12 scientific-computing programs and audit the resulting 60-cell design against default syntactic mutation, relation-pattern alignment, boundary cases, an adjoint extension arm, and a compact real-defect evidence slice. The evidence shows that semantic alignment, aggregate mutant kill-rate, and real-defect detection are related but distinct constructs. SMS should therefore be read as a construct-level adequacy metric for declared semantic strata, not as a universal dominance claim over existing mutation scores.

CCS Concepts: • **Software and its engineering → Software testing and debugging**; *Software verification and validation.*

Additional Key Words and Phrases: metamorphic testing, mutation testing, semantic mutation operators, metamorphic relation adequacy, LLM-generated mutants, scientific computing

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1 Introduction

1.1 Motivation and scope

Metamorphic Testing (MT) addresses the test-oracle problem in scientific computing software: instead of checking outputs against a known-correct reference, MT checks metamorphic relations (MRs), invariants connecting outputs of related inputs. While *semantic mutation testing* has been named since Clark et al. [11] and tooled for C [16] and for UML state-machine behaviour [19], and while MR-adjacent data-semantic mutation [1, 2, 10, 43] and recent LLM semantic-invariance MT [14] extend the mutation target to test data, relations, and task-level semantics, the adequacy of an MR set against *domain-specific code-level faults*, conservation laws, monotonicity, convergence order, trajectory

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shape, fidelity ordering, has lacked a backward-compatible metric tied to the classical Mutation Score (MS) lineage. Jia and Harman's [2011] MS is defined over syntactic Abstract Syntax Tree (AST) mutations and does not capture these domain semantics. Recent Large Language Model (LLM) generated mutants [26, 37] do produce semantically richer mutants, but do not separate the contribution of LLM source diversity from the contribution of MR design.

Throughout this paper we use the term *four representative classes of single-output scientific computing kernels* (numeric, probabilistic, surrogate, and machine-learning), and avoid the ambiguous software-engineering term *paradigm*. The scope of our claims is strictly bounded to single-output `float` → `float` kernels: each Program Under Test (PUT) is under 2 KB. We do not claim industrial transfer for multi-output software in this paper.

A compact real-defect evidence slice then checks the same semantic distinction developed here: mutation kill-rate, semantic stratum alignment, and real-defect detection are related but different constructs. The slice strengthens the semantic-mutant argument at result level; it is not presented as a reusable benchmark artifact. The slice was selected only after defect/MR detectability verification and is reported here at result level. Candidate mining, rejected cases, repository tooling, and benchmark release are outside this manuscript.

1.2 Three-layer methodological framework

We organise the contribution as a three-layer framework around domain-semantic mutation operators.

- **Layer 1, Definitional (§3.2).** A mutation is *semantic* if it satisfies at least one of three necessary conditions: (a) it crosses a function-call or module-import boundary, (b) it depends on domain knowledge for legality, or (c) it changes the algorithmic class. The five meta-operator classes, Conservation Erosion (CE, also written mut_C), Operator Substitution (OS, mut_M), Hyperparameter (HP, mut_G), Trajectory Flip (TF, mut_T), and Structural Injection (SI, mut_F), specialise these conditions across the four PUT classes.
- **Layer 2, Operational (§2.3, §3.10).** An equivalence judgement E1 ∧ E2, where E1 is Automated Verification Pipeline (AVP) coherence and E2 is output equivalence on K_eq = 1000 samples, gives the conservative instantiation of the Layer-1 conditions. The trade-off against E1-alone and E2-alone variants is in Appendix A.3.
- **Layer 3, Applied (§3.5).** AST-normalised empirical traceability across all 12 PUTs: comparing 292 v4 mutants against 1,250 cosmic-ray syntactic mutants, we show empirically that the v4 pool is not a subset of the syntactic-mutant pool.

The 60-cell empirical audit reported in Section 4 stress-tests this backbone. SMS is positioned as a strict generalisation of classical MS: under the degenerate limit `L = L_equiv ∧ L_killed ∧ L_mut` formalised in §2.6, SMS reduces almost everywhere to MS. Layer 1 and Layer 3 correspond to the two leading research questions, RQ1 (constructibility and backward compatibility) and RQ2 (syntactic unreachability); the 60-cell demonstration answers RQ3–RQ5.

The empirical evidence has four roles. The 12-PUT audit is the main stress test of the SMS construction. The AST audit tests whether the semantic-mutant space is reachable by default first-order syntactic mutation. The boundary and adjoint arms map when strong MRs do, and do not, detect semantic effects. The real-defect slice checks whether the construct distinctions remain visible on verified MR-detectable defects.

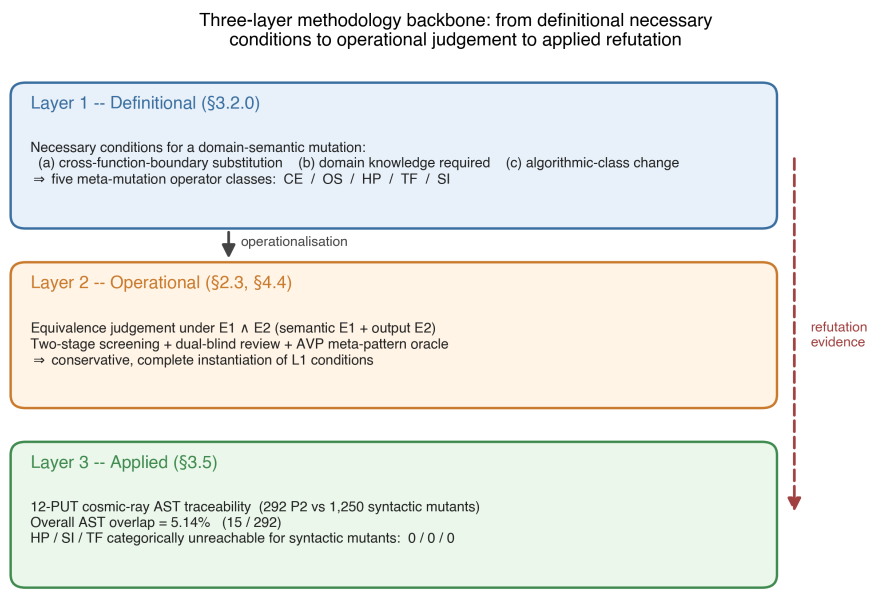

Fig. 1. Three-layer methodological framework: Layer 1 (definitional necessary conditions), Layer 2 (operational $E_1 \wedge E_2$ judgement), Layer 3 (applied AST-normalised traceability across 12 PUTs).

Table 1. Claim-evidence map.

Claim	Evidence	Verdict	Scope
SMS is backward-compatible with MS	Theorem 9.1 and Corollary 9.1	Supported	Degenerate syntactic limit
First-order syntactic coverage is limited	5.14% overlap; HP/SI/TF at 0/0/0	Supported	cosmic-ray defaults; higher-order mutation not refuted
Operator-MP alignment produces a large SMS effect	H2 threshold test	Not met	12-PUT stress test; exploratory
SMS diagnoses MR adequacy	LRCA annotations, boundary cases, and real-defect slice	Supported, qualified	Small PUTs plus narrow real-defect slice
Real-defect detection follows mutant kill-rate	Real-defect face	Rejected	Constructs separate

This table also marks the paper's non-claims. We do not claim that SMS is universally superior to classical MS, that pattern-derived MRs dominate all generic relations, that higher-order mutation has been refuted, that the real-defect slice is a benchmark contribution, or that the reported SMS values define industrial acceptance thresholds.

1.3 Related work

Seven lines of prior work bracket the contribution of this paper.

(a) Classical mutation testing. Jia and Harman [30] and Papadakis et al. [34] survey syntactic mutation. The Coupling Effect Hypothesis (CPH), that tests detecting simple faults also detect complex faults, underpins the use of MS as a fault proxy [6, 18, 31]. We argue in §3.5 that CPH holds within syntactic mutation but does not automatically extend across the syntactic-versus-domain-semantic boundary, even when it couples simple syntactic faults to complex syntactic faults. The §3.5 empirical evidence (5.14% AST overlap on 12 PUTs, with HP, SI, and TF at 0/0/0) is the empirical witness for this boundary.

(b) Higher-order mutation. Jia and Harman [29] and Kintis et al. [32] study compositions of first-order syntactic mutants. We explicitly exclude any equivalence claim about Higher-Order Mutation (HOM) and list HOM as residual threat R12 (Appendix F.2).

(c) LLM-generated mutants. Tip et al. [37] introduce LLMorpheus, which uses single-LLM JavaScript mutants. Humbatova et al. [26] introduce DeepCrime for deep-learning real-fault mutation. Dakhel et al. [15] extend LLM-driven mutation testing to test generation on Java PUTs, providing an *Information and Software Technology* anchor for the LLM-mutant lineage. To our knowledge no prior LLM-mutant work isolates *LLM source diversity* from *MR design* in the contribution to effect size; the three-stage ablation in §3.9 closes this gap. On the equivalent-mutant problem, Delgado-Pérez and Chicano [17] emphasise the importance of the `equiv` term in the MS denominator; we extend that classical bitwise-equivalence definition to a semantic-class equivalence $E1 \wedge E2$ in §2.3. Zhang et al. [42] demonstrate MT validation in class-integration test ordering, a non-scientific-computing domain, and the present paper specialises MT to scientific-computing PUTs and couples it to a domain-semantic mutation operator framework.

(d) Semantic mutation operators across testing communities. Semantic mutation as a research line is older than the LLM-mutant lineage. Clark et al. [11] introduce *semantic mutation testing* as a path distinct from syntactic mutation; Dan and Hierons [16] tool the approach for C; Derezińska and Zaremba [19] apply *semantic mutation operators* to UML state-machine behaviour by perturbing semantic-variation-point interpretations rather than syntactic structure. Within MT specifically, ai Sun et al. [1, 2] propose *data mutation operators* and the μMT methodology to drive MR acquisition through controlled data-semantic perturbation; Zhu et al. [43] formalise *datamorphic testing*, separating datamorphisms (test-case-to-test-case mappings) from metamorphisms (test-case-to-Boolean mappings) over data semantics; Chan and Keung [10] extend generic data mutation operators to unsupervised software-defect-prediction validation. Most recently, Curtò and Zarzà [14] formalise *semantic invariance* in LLM MT under paraphrase, fact-reorder, expand/contract, and context-shift transformations. These works establish that mutation can target data, relation, task, and behavioural semantics. **The contribution of this paper is the first to formalise a domain-specific Semantic Mutation Score (SMS) for MR adequacy in scientific computing kernels, with a proven backward-compatible reduction to classical MS (Theorem 9.1) and a code-level operator framework (Conservation Erosion, Operator Substitution, Hyperparameter, Trajectory Flip, Structural Injection) targeting kernel-specific faults, and to delineate the strong boundary of metamorphic-relation grounding, a strong/weak MR distinction whose ε_{tol}-governed duality (Theorem 2, Proposition 2) is mapped empirically on real and textbook programs spanning the meta-patterns (§4.8), complemented by a cross-library adjoint extension arm (§3.7, §4.9) that supports the duality forward on third-party linear operators with real fixed defects.**

Table 2. Closest denominator-oriented comparators to SMS. The comparison is by adequacy object and denominator semantics, not by whether a paper uses the phrase metamorphic-testing adequacy.

Work	Denominator or counted object	Relation to SMS
[21, 33]	MR obligations, MR coverage, and source-test obligations	Closest MR/MT adequacy comparators, but property/coverage denominators rather than semantic mutant classes
[7]	Differentially executed code elements in paired MT runs	Execution-side coverage comparator, not MR-set semantic adequacy
[40]	MR feature/diversity space	Boundary surrogate for MR-set quality, not an adequacy fraction over faults or semantic classes
[25, 41]	MR validity, coverage, and classical mutation-score signals	Boundary surrogate for cost-effective MR choice, not a denominator-bearing adequacy metric
[9]	Semantic obligations under a specification	Non-MT semantic-coverage analogue; test-suite side rather than MR-set side

(e) MR adequacy and coverage. Recent MT-specific adequacy work has begun to ask how MR-based tests should be measured rather than merely generated. The closest literature is best separated by its *counting object*. Xiang et al. [39] construct composite MRs for GSL scientific functions, but evaluate them with the classical MuJava denominator of nonequivalent syntactic mutants. Fu et al. [21] define denominator-bearing MT adequacy over MR obligations and source inputs; Liu et al. [33] integrate MR coverage into MT adequacy; and Ba et al. [7] define Metamorphic Coverage over code elements differentially executed by paired MT runs. These are the closest denominator-bearing MR/MT coverage comparators, but their denominators are MR obligations, source-test obligations, or MR-induced execution coverage rather than semantic mutant classes. Xie et al. [40] quantify MR-set diversity with the MUT model, while Huang et al. [25] and Zhang et al. [41] rank or select MRs using validity, coverage, or mutation-score signals. These latter works are important boundary surrogates for MR-set quality, but they do not define an adequacy fraction over an enumerated universe of semantic fault classes. Blwi et al. [9] provide a useful non-MT analogue by defining test-suite effectiveness through semantic coverage, showing that denominator design can move beyond syntax; however, their adequacy object is a test suite under a specification, not an MR set for scientific kernels. Thus, prior work measures MR exercise, input coverage, differential execution, MR diversity, or syntactic-mutant detection. None makes the adequacy denominator the set of admitted nonequivalent *domain-semantic code mutants* for scientific kernels, and none proves compatibility with the classical MS denominator. SMS therefore differs not by proposing "another MR coverage metric", but by replacing the denominator against which an MR set is judged.

(f) Property-relative and numerical-specification mutation. Recent property-aware mutation work moves the adequacy question closer to requirements and invariants: Bartocci et al. [8] make mutant relevance and meaningful killing relative to a tested property, and Jeangoudoux et al. [27] generate tests for numerical specifications through interval-constraint reasoning over floating-point accuracy and robustness requirements. These works are closer to SMS than generic AST mutation because they reject property-irrelevant mutants as an adequacy denominator. They still differ from the present paper in three respects: the adequacy object here is an MR set rather than a single formal requirement, the denominator is a five-class domain-semantic mutant space for scientific kernels, and the metric is constructed as a backward-compatible generalisation of MS.

(g) Semantic foundations and the certificate gap. Abstract interpretation supplies the standard machinery for relating concrete semantics to abstract domains and invariant families [12], and for designing program transformations by their effect on an abstract semantics [13]. Approximate program transformation semantics gives a second, quantitative route: local error expressions and partial-metric semantics can certify that a transformation lies within a bounded semantic-effect envelope [22, 38]. Refinement- and UTP-based mutation testing lift mutation to contracts and specifications [3, 4], while usage-aware differential dynamic logic tracks admissible proof-level mutations under soundness conditions [20]. These foundations supply much of the machinery needed for certificate-bearing semantic-effect classes, but they are not MR adequacy metrics for scientific-computing kernels. Conversely, SMS provides the MR adequacy denominator and empirical audit, but its certification layer is empirical and structural rather than machine-checkable proof-carrying.

Closest comparison. The closest adjacent work is the Min-MR-Complete formulation (Li et al., 2026, arXiv preprint), which uses the same semantic-mutation family as the fault model for MR-subset completeness and identifies when class-level reasoning collapses to a mutant-level set-cover problem. The present paper differs in role: it defines the SMS denominator, the E1 $\wedge$ E2 admission judgement, the AST-normalised syntactic-unreachability audit, and the empirical same-source/cross-source ablation. Min-MR-Complete optimises over an admitted mutant–draw coverage universe; this paper constructs and audits that universe. Taken together, the two papers come closest to semantic-effect-class denominators for MR adequacy, but neither yet attaches a per-mutant machine-checkable certificate showing that a syntactic edit belongs to a declared abstract semantic-effect class.

The SMS metric is conceptually complementary to the code-verification scope of ASME V&V 20-2009 §3, which targets numerical-solver correctness. SMS targets MR fault-detection adequacy; we make no normative compliance claim (Appendix E.3 records this as a long-term aspiration only).

1.4 Research questions

The research questions are ordered from the theoretical foundation (RQ1), through the structural separation that motivates it (RQ2), to the empirical demonstration on the 60-cell audit (RQ3–RQ5). RQ1 is answered by construction and proof (§2.7–2.9: Theorem 1 undecidability, Proposition 1 fiber characterisation, and Theorem 2 duality, together with the SMS–MS degeneration of §2.6); RQ2 by the AST-normalised structural audit (§3.5); RQ3–RQ5 by the pre-registered statistical pipeline (§4). This ordering mirrors the three-layer framework of §1.2: Layer 1 (definitional) carries RQ1, Layer 3 (applied) carries RQ2.

- **RQ1 (Theory of semantic mutation: constructibility, undecidability, duality, backward compatibility).** Can semantic mutation be formalised as a fiber decomposition of a single effect map over admitted nonequivalent *domain-semantic code mutants* for scientific-computing kernels, such that (a) stratum membership is undecidable and therefore certificate-based (Theorem 1), (b) the induced adequacy metric SMS degenerates almost everywhere to the classical Mutation Score (Proposition 1; Theorem 9.1), and (c) strong metamorphic relations detect only the semantically active mutants within a tolerance-bounded regime, while outside it the duality degrades along an ε_{tol}-governed strong boundary into false positives (weak MRs, where a correct discrete program violates the invariant) and false negatives (surviving non-equivalent mutants under coincidental satisfaction) (Theorem 2 and Proposition 2; the boundary is mapped empirically under RQ4)?
- **RQ2 (Syntactic unreachability).** To what extent is the domain-semantic mutant space induced by the five operators reachable by default-configuration first-order syntactic mutation tools at the AST-normalised level?
- **RQ3 (Instantiability).** Do the five semantic mutation operators instantiate a usable nonequivalent semantic-mutant denominator across the 60 PUT-MP-operator cells, as reflected by inst_rate, equiv_rate, C1_share, and survive_rate?

- **RQ4 (Discrimination, LLM-source decoupling, and the strong boundary).** Does SMS discriminate operator-MP aligned slices (j=k) from cross slices (j≠k); does cross-source LLM pooling under an identical prompt move the aligned-versus-cross effect size, or only the kill-set quality; and where does the strong boundary of Proposition 2 lie empirically, on which real and textbook programs does a strong MR kill the genuine fault, and where does the duality fail as a weak-MR false positive or as a surviving non-equivalent mutant?
- **RQ5 (Cross-class consistency).** Does the aligned-versus-cross SMS pattern generalise across four representative classes of single-output scientific-computing kernels? (The relationship between SMS and the coverage-style surrogate Pattern Coverage is reported descriptively at n = 12 as a hypothesis-generating observation, not a confirmatory test.)

1.5 Hypotheses

The empirical questions RQ3–RQ5 carry pre-registered hypotheses; RQ1 is answered by Theorem 9.1 and RQ2 by the §3.5 structural audit, neither of which is a statistical hypothesis.

- **H1 (RQ3).** At least 4 of the 5 operators produce ≥ 5 non-equivalent mutants on at least 9 of the 12 PUTs.
- **H2 (RQ4; pre-registered, exploratory).** The aligned-SMS to cross-SMS odds ratio is ≥ 3.0, and Cliff's δ is ≥ 0.474. This large-effect threshold is an underpowered exploratory target (point-estimate verdict and stipulated-alternative power in §4.4), not a confirmatory claim of the paper.
- **H3 (RQ5).** A within-class sign test gives 4/4 across the 4 classes, and CV(ΔSMS) < 0.5.
- **H4 (RQ4).** Mean suspect_share is ≤ 0.20 across the 60 cells.

The 12 PUTs are deliberately *Numerical Recipes*–style compact kernels: they provide a verifiable minimum working example for the three-layer backbone, while industrial-scale transfer is left to future work (Threat R5).

2 Theory: SMS metric and semantic mutation

2.1 Minimal vocabulary and symbols

We use a small vocabulary throughout the method. The terms are declared here before they are used in the SMS construction.

Program Under Test (PUT). A PUT is the scientific-computing kernel being assessed. We write the i-th PUT as S_i.

Metamorphic Relation (MR). An MR is a property that relates the outputs of related inputs. The MR family used for PUT S_i and meta-pattern k is written $\mathrm{MR}_{i,k}$.

Meta-pattern (MP). An MP is a semantic stratum of MR behaviour, such as conservation, monotonicity, convergence, trajectory structure, or fidelity order. The index k denotes the MP being tested.

Semantic operator. A semantic operator is a mutation family mut_j designed to perturb a declared MP. The operator index j identifies the intended semantic effect.

Semantic mutant. A semantic mutant s' is a finite syntactic edit of S_i that is admitted by semantic operator mut_j and is certified against the declared MR family. Formally,

$$s' \in \mathrm{mut}_j(S_i).$$

The word semantic therefore names the certification target, not a second editing mechanism.

Semantic-effect fiber. For an MR family $\mathrm{MR}_{i,k}$, the semantic-effect fiber is the subset of admitted semantic mutants whose declared effect is observable by that MR family:

$$\mathcal{F}(\mathrm{MR}_{i,k}) = \{s' \in \mathrm{mut}_j(S_i) \, : \, \mathrm{killed}(s', \mathrm{MR}_{i,k})\}.$$

Different MR families can observe different fibers. SMS therefore measures adequacy relative to a declared stratum; it is not a total order over all possible MR families.

Equivalent, killed, and surviving mutants. A mutant is equivalent for (i, k, j) when the equivalence judgement of §2.3 admits it into $\mathrm{equiv}_{i,k,j}$. A non-equivalent mutant is killed when at least one MR in $\mathrm{MR}_{i,k}$ passes on S_i and fails on s'. A non-equivalent mutant that is not killed is surviving. Thus the generated mutant set decomposes as:

$$\mathrm{mut}_j(S_i) = \mathrm{equiv}_{i,k,j} \cup \mathrm{killed}_{i,k,j} \cup \mathrm{survive}_{i,k,j} \quad \text{(disjoint)}$$
$$\mathrm{SMS}_{i,k,j} := \frac{|\mathrm{killed}_{i,k,j}|}{|\mathrm{mut}_j(S_i)| - |\mathrm{equiv}_{i,k,j}|} \in [0,\ 1]$$

Semantic Mutation Score (SMS). SMS preserves the classical Mutation Score (MS) ratio while replacing the internal definitions of mutant generation, equivalence, and killing with MR-relative semantic definitions. The classical concepts of mutation operator, mutant, equivalent mutant, killed mutant, surviving mutant, and MS are inherited from Jia and Harman [30] and Ammann and Offutt [5].

The complete notation table is in Appendix A.1; index sets and tolerance parameters ($\varepsilon_{\mathrm{eq}}, \varepsilon^k_{\mathrm{AVP}}, K_{\mathrm{eq}} = 1000, N = 20$) follow that table.

2.2 Operator signatures

Operator family and alignment:

$$\mathrm{MUT} = \{\mathrm{mut}_C,\ \mathrm{mut}_M,\ \mathrm{mut}_G,\ \mathrm{mut}_T,\ \mathrm{mut}_F\} \quad \text{(open, extensible)}$$
$$\mathrm{mut}_j \, : \, \mathrm{Programs} \to 2^{\mathrm{Programs}}$$
$$\mathrm{align}(j) = j \quad \text{(design choice, not theorem)}$$

Table 3. Operator signatures.

Operator	Failure semantics	Aligned MP
mut_C	Conservation-breaking	MP_1 Conservation
mut_M	Monotonicity-breaking	MP_2 Monotonicity
mut_G	Convergence-breaking	MP_3 Convergence
mut_T	Trajectory-distorting	MP_4 Trajectory
mut_F	Fidelity-order-breaking	MP_5 Partial-order

Per-PUT specialisation tables (mut_C/M/G/T/F across 12 PUTs and 4 classes) are deferred to **Appendix B.2**.

2.3 Equivalence judgement E1 and E2

The equivalence judgement is the executable instantiation of the Layer-1 necessary conditions (§3.2.0). For each candidate mutant `s'`:

- **E1 (AVP-coherent):** ∀ mr ∈ $\mathrm{MR}_{i,k}$: $\mathrm{AVP}(S_i, mr) = \mathrm{AVP}(s', mr)$.
- **E2 (Output-equivalent):** ∀ x in $K_{\mathrm{eq}} = 1000$ samples $\sim D_S$: $\|S_i(x) - s'(x)\| \leq \varepsilon_{\mathrm{eq}}$.

E1 ∧ E2 is the conservative complete instantiation: false-equiv (a truly non-equivalent mutant admitted as equivalent) requires *both* AVP coherence and numerical agreement on K_eq samples to hold despite the non-equivalence (both mis-pass simultaneously), which is rare; false-non-equiv requires only one of E1, E2 to fail for a truly equivalent mutant, biasing SMS slightly high. The trade-off table against E1-alone and E2-alone is in Appendix A.3. Under the §2.6 degenerate limit `L`, E1 ∧ E2 reduces almost everywhere to classical bitwise equivalence (Lemma 9.1).

The killed determination preserves the classical OR-aggregation:

$$\mathrm{killed}(s', \mathrm{MR}_{i,k}) \iff \exists\, mr \in \mathrm{MR}_{i,k} : \\ \mathrm{AVP}(S_i, mr) = \mathrm{pass} \wedge \mathrm{AVP}(s', mr) = \mathrm{fail}.$$

2.4 Likely Root-Cause Analysis diagnostic layer

Likely Root-Cause Analysis (LRCA) annotates every killed mutant with one of five diagnostic labels: C1 likely semantic failure, C2 numerical-tolerance perturbation, C3 out-of-distribution (OOD) trip, C4 statistical-assumption violation, and C5 mutator artefact. The diagnostic procedure is a three-layer decision tree. Layer 1 checks tolerance robustness over N = 20 repeats with a fail-ratio cutoff of 0.80. Layer 2 triages OOD on the surrogate and machine-learning classes. Layer 3 checks statistical-assumption baselines on the probabilistic and machine-learning classes using Wilcoxon and Dynamic Time Warping. A final pass rechecks for mutator artefacts. Multiple labels are resolved by priority C5 > C4 > C3 > C2 > C1.

The output quantities are `C1_share` (share of C1 among killed mutants) and `suspect_share := 1 − C1_share`. **LRCA does not modify the SMS formula**: the killed set is never filtered by suspect status. LRCA is a diagnostic annotation that tells the reader whether a given kill is more consistent with a semantic-failure detection, tolerance sensitivity, OOD behaviour, statistical-assumption failure, or mutator artefact. Appendix A.2 gives the full decision tree, the 9-grid threshold calibration, and the engineering rationale for the priority ordering.

2.5 Backward-compatibility declaration

The seven core concepts (PUT, mutation operator, mutant, equivalent mutant, killed mutant, surviving mutant, mutation score) and the SMS formula `|killed| / (|mut| − |equiv|)` are aligned item-for-item with Jia and Harman [30]. We extend only the internal definitions: `mut` shifts from syntactic to domain-semantic operators, `equiv` shifts from bitwise behavioural equivalence to the semantic-class equivalence E1 ∧ E2, and `killed` shifts from an equality oracle to a meta-pattern AVP. We introduce no new formula terms and no new state classifications.

Under the degenerate limit `L` defined in §2.6, SMS reduces almost everywhere to classical syntactic MS, equivalent mutants degenerate to classical behavioural equivalence, killed mutants degenerate to bitwise difference detection, and the LRCA layer trivialises, C2 through C5 cannot fire when L1 ∧ L2 ∧ L3 hold. Any SMS-based empirical conclusion is therefore structurally consistent with the existing mutation-testing literature in the classical syntactic-mutation regime, so there is no metric-level semantic fragmentation. The skeleton of eleven concepts, the three-state decomposition, the SMS formula, and the AVP interface are fixed; the contents of MUT, MP, C, and `cls` remain open to future extension.

2.6 SMS-to-MS degeneration: formal statement

The §2.5 backward-compatibility claim is formalised as a degeneration theorem. The full notation cross-reference is **Appendix G.1**; the joint conditions and lemma proofs are in **Appendix G.2-G.4**. We state only the main theorem and corollary here. Theorem and lemma labels (Theorem 9.1, Corollary 9.1, Lemma 9.1-9.3) are preserved as stable identifiers for cross-reference with Appendix G.

Definition (Degenerate limit). $L = L_{\text{equiv}} \wedge L_{\text{killed}} \wedge L_{\text{mut}}$, where each joint condition is a pair of paired axes acting on one layer of the SMS formula:

- $L_{\text{equiv}} = L1 \wedge L2$: $\varepsilon_{\text{eq}} \to 0 \wedge K_{\text{eq}} \to \infty$ (controls equiv layer; Lemma 9.1).
- $L_{\text{killed}} = L3 \wedge L4$: $\varepsilon_{\text{AVP}} \to 0 \wedge$ MP set = $\{\text{MP}_{\text{eq}}\}$ (controls killed layer; Lemma 9.2).
- $L_{\text{mut}} = L5 \wedge L6$: mut_j switches to rule-based syntactic operators (Mothra-style, or modern MutMut/PIT-style) $\wedge$ PUT class $\subseteq$ imperative deterministic programs (controls mut layer; Lemma 9.3).

Theorem 9.1 (SMS → MS degeneration). In the degenerate limit `L`, **almost everywhere** with respect to the input distribution `D_S`,

$$\text{SMS}_{i,k,j} \xrightarrow{L} \text{MS}_{i,j} = \frac{|\text{killed}_{i,j}^{\text{classic}}|}{|\text{mut}_j^{\text{syntax}}(S_i)| - |\text{equiv}_{i,j}^{\text{classic}}|}$$

where the right-hand side is the classical Mutation Score of Jia and Harman [30], `killed^classic` is the difference-detection set, `equiv^classic` is the behavioural-equivalence set, and `mut^syntax` is the syntactic-mutant set.

Proof sketch. By Lemma 9.1, equiv_{i,k,j} → equiv_{i,j}^classic almost everywhere with respect to D_S; the measure-zero qualifier accommodates floating-point pathological points and Not-a-Number propagation. By Lemma 9.2, killed_{i,k,j} → killed_{i,j}^classic, and the MP index k degenerates because L4 collapses MR_{i,k} to {MP_eq}. By Lemma 9.3, mut_j(S_i) → mut_j^syntax(S_i). Substituting into the SMS formula gives MS_{i,j}. Full lemma proofs are in Appendix G.3 and the full theorem proof in Appendix G.4.

Corollary 9.1 (LRCA trivialisation). Under `L`, the likely-root-cause inventory C = {C1, ..., C5} degenerates to {C1}, so suspect_share → 0 and SMS becomes a single-layer metric consistent with the engineering attribution structure of Jia and Harman [30].

Empirical consistency. Theorem 9.1 + Corollary 9.1 jointly guarantee that any SMS-based empirical conclusion (§4 Cliff's delta, Friedman chi^2, Spearman rho) is structurally consistent with existing mutation-testing literature in the classical syntactic-mutation regime, and **does not** constitute metric-level semantic fragmentation.

2.7 A theory of semantic mutation: setting

Sections 2.1–2.6 fix the SMS metric and its degeneration to MS. We now give the theory the metric instantiates: semantic mutation is *intensionally semantic, extensionally syntactic*. Every mutation is mechanically a syntactic act, because semantics changes only through syntax. What makes a mutant *semantic* is where its defining condition lives (at the semantic layer, as a specified invariant violation) and how its membership is established (by certificate, because the condition is undecidable).

Fix a grammar G and the set $\mathrm{Prog}(G)$ of well-formed programs with denotational semantics $\llbracket \cdot \rrbracket : \mathrm{Prog}(G) \to (\mathcal{X} \rightharpoonup \mathcal{Y})$. Let $\alpha : \mathcal{Y} \to \mathcal{Y}_\alpha$ be the committed semantic abstraction (the observable that the AVP of §2.3 reads), and write $P \equiv_\alpha P'$ when $\alpha \circ \llbracket P \rrbracket = \alpha \circ \llbracket P' \rrbracket$ pointwise on the admitted input domain $\mathcal{X}_{\mathrm{adm}}$. This $\equiv_\alpha$ is exactly the equivalent-mutant relation equiv of §2.1: the E2 component of the E1∧E2 judgement (§2.3) is its bounded decision procedure on K_{eq} samples.

The new ingredient is a finite *semantic invariant family* $I = \{\psi_1, \dots, \psi_5\}$, where each ψ is a predicate over the observable behaviour $\alpha \circ \llbracket P \rrbracket$. The five invariants are exactly the properties the meta-patterns of §2.2 target: ψ_1 conservation, ψ_2 monotonicity, ψ_3 convergence order, ψ_4 trajectory determinism, ψ_5 partial-order consistency, so ψ_k is the invariant whose violation MP_k detects and whose breakage mut_j (at $j = k$) targets. We write $\llbracket P \rrbracket \models \psi$ when P's observable behaviour satisfies ψ.

The family is open (the extensibility of I noted in §2.9): a sixth invariant ψ_6, *adjoint consistency* $\langle Ax, y \rangle = \langle x, A^{\mathrm{H}} y \rangle$ for linear-operator kernels, instantiates the same construction. Its MR is strong (its violation set is closed under $\equiv_\alpha$), so by Theorem 2 it detects only the semantically active mutants of the adjoint code path. Because adjoint consistency is meaningful only for linear operators , not for the chaotic, stochastic, and nonlinear machine-learning kernels that make up most of the 12 PUTs, ψ_6 is *not* folded into the pre-registered 60-cell design (doing so would leave the invariant undefined on ten of the twelve subjects); it is instead demonstrated as a separate confirmatory *adjoint extension arm* on dedicated third-party linear-operator subjects (§3.7, §4.9), leaving the pre-registered hypotheses H1–H4 untouched.

2.8 Semantic mutant: conditions S1–S5

The following certified definition is the formal counterpart of the informal Layer-1 necessary conditions of §3.2.

Definition (semantic mutant). Given (P, ψ) with $\llbracket P \rrbracket \models \psi$, a program P' is a *semantic mutant of P at stratum ψ* iff: (S1) *realizability*, $P' = e(P)$ for a finite AST edit e; (S2) *baseline*, $\llbracket P \rrbracket \models \psi$; (S3) *violation with witness*, $\llbracket P' \rrbracket \not\models \psi$, witnessed by some $x \in \mathcal{X}_{\mathrm{adm}}$ observable through α; (S4) *latency*, P' compiles, terminates on $\mathcal{X}_{\mathrm{adm}}$, and is type-correct, so the defect is observable only at the semantic layer, never by a crash oracle; (S5) *stratum purity* (required where stratum labels feed downstream), $\llbracket P' \rrbracket \models \psi'$ for all $\psi' \in I \setminus \{\psi\}$.

A semantic mutation class is therefore defined by which invariant must break and how (S2–S5) and realized by whichever syntactic edits achieve it (S1). The operators $\mathrm{mut}_C, \dots, \mathrm{mut}_F$ of §2.2 are the constructive templates for strata $\psi_1, \dots, \psi_5$.

Definition (latency window). For a violation-magnitude parameter ε of an edit template, the *latency window* is the interval $(\varepsilon_{\mathrm{tol}}, \varepsilon_{\mathrm{crash}})$: above the tolerance at which the ψ-checker witnesses the violation (S3), below the threshold at which the container crashes (S4). An empty window certifies non-realizability of the template at that site.

2.9 Effect map, fibers, and three theorems

Definition (effect map and fibers). For fixed P, the *effect map* σ sends an applicable edit e to the semantic-effect class of $P_e = e(P)$, valued in $\{\equiv_\alpha,\ \text{ill-formed},\ \psi_1\text{-viol}, \dots, \psi_5\text{-viol},\ \text{active-off-taxonomy}\}$, where active-off-taxonomy means $P_e \not\equiv_\alpha P$ yet no $\psi \in I$ flips from satisfied to violated. The *fiber* of a class is its σ-preimage. P_e is *semantically active* iff $P_e \not\equiv_\alpha P$; the strict syntactic-only mutants are exactly the inactive fiber $\sigma^{-1}(\equiv_\alpha)$.

Theorem 1 (undecidability of semantic-mutant membership). For Turing-complete $\mathrm{Prog}(G)$ and non-trivial ψ, deciding whether an edit e is a semantic mutant at stratum ψ (conditions S2–S4) is undecidable. *Proof sketch.* S3 requires deciding $\llbracket P_e \rrbracket \not\models \psi$, a non-trivial semantic property, undecidable by Rice's theorem; the termination clause of S4 is independently undecidable by reduction from the halting problem. Semantic-mutant status therefore cannot be read off the edit; it can only be certified on bounded fragments with recorded budgets, which is the formal reason the admission judgement (§2.3, §3.10) is gate-shaped rather than a decision procedure.

Proposition 1 (fiber characterization of classical practice). (i) Every semantic mutant arises from some syntactic edit; the stratum-ψ class equals $\sigma^{-1}(\psi\text{-viol})$ intersected with S4. (ii) The classical equivalent-mutant problem is exactly the degenerate fiber $\sigma^{-1}(\equiv_\alpha)$ of unconstrained syntactic sampling. (iii) Classical mutation testing and semantic mutation are the same effect map under two sampling policies: sample the edit domain blindly, versus sample a specified non-degenerate fiber. *Proof.* (i) holds because semantics changes only through syntax; (ii) and (iii) follow by unfolding the definitions. The SMS-to-MS degeneration of Theorem 9.1 (§2.6) is the metric-level shadow of (ii): when the admitted fiber collapses to the syntactic default, the SMS denominator collapses to the MS denominator.

Theorem 2 (duality with strong MRs). Let r be a metamorphic relation whose violation set is closed under $\equiv_\alpha$ (a *strong* MR). Then r detects no semantically inactive mutant, and conversely any mutant detected by a strong MR is semantically active. Hence over a strong MR universe the kill matrix is supported exactly on the semantically active fibers, and inactive (syntactic-only) mutants are structural noise for strong MT evaluation rather than hard cases. *Proof.* If $P_e \equiv_\alpha P$ and r's violation predicate factors through α, then r flags P_e iff it flags P; MR validity says r does not flag P, so it does not flag P_e. The converse is the contrapositive. □

The duality identifies α as the single hinge connecting MR-side quality (strength) and mutant-side quality (activity). Classification is relative to $(I, \alpha, \text{budget})$: extending I can only move a mutant from active-off-taxonomy to ψ-stratified, never the reverse, and bounded $\equiv_\alpha$ verdicts are search conclusions, not proofs, so every certificate carries its $(I, \alpha, \text{budget})$ stamp. The full reduction for Theorem 1 follows Rice's theorem together with the standard halting reduction for the termination clause; the proof of Theorem 2 is the $\equiv_\alpha$-closure argument above.

Definition (strong MR, weak MR, strong boundary). Theorem 2 reads off strength algebraically, assuming the invariant ψ behind r is satisfied by the correct program so that closure under $\equiv_\alpha$ makes a violation diagnostic. For a discrete program P approximating a continuous operator this assumption is *contingent*: the correct P reproduces the operator's algebraic invariant only up to a discretisation residual, so $\equiv_\alpha$ between P and the ideal solution holds only within the tolerance ε_{tol} of the latency-window definition. Fixing ε_{tol}, call r *strong* on P when the correct P satisfies r within ε_{tol} (its residual stays below ε_{tol} under refinement), so that any violation beyond ε_{tol} certifies a semantically active mutant and Theorem 2 holds operationally; call r *weak* on P when the correct P's residual exceeds ε_{tol}, so that the correct program is itself flagged because the discretisation, not a fault, breaks the invariant. The *strong boundary* is the $(\varepsilon_{\text{tol}}, \text{resolution})$ locus separating the two regimes; tightening ε_{tol} or coarsening the discretisation carries a strong MR across it into the weak regime. A weak MR is exactly a non-strong (non-grounded) MR in the effective, discrete sense: its violation set is not closed under the program's ε_{tol}-level $\equiv_\alpha$.

Proposition 2 (the two boundary failures and their ε_{tol} coupling). Relative to $(\varepsilon_{\text{tol}}, P)$ the duality of Theorem 2 has two failure modes. (i) *False positive (weak MR).* When r is weak the correct P violates r and a non-fault is reported; the rotational-symmetry invariant of a neutron-transport operator under a method-of-characteristics discretisation, satisfied only at special angles, is the canonical instance. (ii) *False negative (surviving non-equivalent mutant).* A semantically active mutant $P_e \not\equiv_\alpha P$ can leave every invariant in I satisfied within ε_{tol}, a *coincidental satisfaction of the MR*, and so survive the strong MR family; this surviving mutant is non-equivalent (genuinely $\not\equiv_\alpha P$) and is therefore distinct from the classical equivalent mutant of Proposition 1(ii), and when its signature lies below ε_{tol} it is a *tolerance fault*. The two modes are coupled through ε_{tol}: tightening it shrinks the false-negative set (more surviving mutants are killed) but grows the false-positive set (more correct programs are flagged), and loosening it does the reverse. The boundary is therefore an ε_{tol}-governed *sensitivity–specificity trade-off*, not a defect to be engineered away. *Proof.* (i) and

(ii) instantiate the Definition: weakness places the correct program's residual above $\varepsilon_{\mathrm{tol}}$, and coincidental satisfaction places the mutant's signature below it; monotonicity of both sets in $\varepsilon_{\mathrm{tol}}$ holds because both membership predicates are threshold comparisons against the same $\varepsilon_{\mathrm{tol}}$. □

Theorem 2 thus holds within the strong regime, and the $\varepsilon_{\mathrm{tol}}$-governed strong boundary, where a strong MR degrades into a weak MR (false positives) or is evaded by a surviving non-equivalent mutant (false negatives), is mapped empirically on real and textbook programs in §4.8.

3 Study design and methods

3.1 PUT selection

Table 4. PUT selection.

Class	PUT	Name	Mathematical structure	LOC
A Numeric	A1	Lorenz ODE integration	Nonlinear ODE	~150
	A2	LU decomposition	Linear algebra	~80
	A3	FDM 1D heat conduction	Parabolic PDE	~200
B Probabilistic	B1	Beta-Binomial conjugate	Analytic posterior	~60
	B2	MCMC Metropolis-Hastings	Markov chain	~250
	B3	Monte Carlo integration	Importance sampling	~100
C Surrogate	C1	Gaussian Process Regression	Kernel methods	~300
	C2	Polynomial Chaos Expansion	Orthogonal basis	~250
	C3	Neural-net surrogate	MLP substitution	~400
D ML	D1	Multi-Layer Perceptron	Backprop.	~350
	D2	Support Vector Machine	Convex optimisation	~200
	D3	Logistic Regression	Maximum likelihood	~120

The 12 PUTs are taken from the shared 12-PUT infrastructure (Li et al., arXiv preprint) but independently justified along four dimensions: library-stack coverage (numpy 2.4.4, scipy 1.17.1, scikit-learn 1.8.0); mathematical-structure coverage (8 of 12 chapters of *Numerical Recipes*); overlap with existing mutation-testing benchmarks (DeepCrime, Defects4J, and the mutmut and cosmic-ray demos); and signature-simplification trade-offs. The full coverage argument is in Appendix B.1. The `program(x: float) → float` signature is a substantive constraint (§6) that bounds the upper limit of mutant semantic complexity; transfer to industrial multi-output PUTs is left to outside the evaluated scope of this paper.

3.2 Necessary conditions for semantic mutation

Definition (Semantic mutation criteria). A mutant `s' = mut_j(S_i)` is a *semantic* mutation if and only if it satisfies at least one of the following three conditions.

(a) **Cross-function-boundary replacement.** The AST node operated on crosses at least one function-call or module-import boundary. For example, `np.linalg.det(M)` → `np.sum(np.diag(M))`.

(b) **Carries domain knowledge.** The legality of the mutation depends on mathematical, physical, or statistical knowledge of the program's domain, not on purely syntactic type preservation. For example, changing the Gaussian Process Regression `noise_level` from 1e-4 to 1e-1.

(c) **Changes algorithmic class.** The mutation alters the algorithmic class implemented. For example, replacing RK4 with Euler changes the integration order.

A mutation that satisfies none of (a) – (c) is purely *syntactic*: AST-local, domain-agnostic, and class-preserving. The five meta-operator classes (CE, OS, HP, TF, SI) are specialisations of (a) – (c):

Table 5. Necessary conditions for semantic mutation.

Operator class	(a)	(b)	(c)	Primary condition
CE Constant perturbation	×	△	×	partial (b); weakest
OS API replacement	✓	✓	△	(a)+(b)
HP Hyperparameter	×	✓	△	(b)+partial(c)
TF Numerical transform	△	✓	✓	(b)+(c)
SI / CF Structural injection	△	✓	✓	(b)+(c)

Only **CE partially satisfies** the necessary conditions (it is a semantic / syntactic boundary class); OS, HP, TF, SI strongly satisfy at least one of (a), (b), (c).

3.3 60-cell instantiation matrix

Table 6. 60-cell instantiation matrix: per-PUT MR-coverage density across the five MPs (12 PUTs × 5 MPs). •• substantial (30 cells), • moderate (24 cells), ∘ vacant (6 cells; no MR is exercised).

PUT	MP_1 cons.	MP_2 mono.	MP_3 conv.	MP_4 traj.	MP_5 p-ord.
A1 Lorenz	••	•	••	••	∘
A2 LU	••	∘	•	•	••
A3 FDM	••	•	••	••	∘
B1 BetaBin	••	•	∘	∘	•
B2 MCMC	•	••	••	••	•
B3 MC	••	∘	••	•	∘
C1 GPR	•	••	••	•	••
C2 PCE	••	•	••	•	••
C3 NN-Surr	•	••	••	••	••
D1 MLP	••	••	•	•	••
D2 SVM	•	••	•	∘	••

D3 LR	••	••	•	∘	••

Legend: •• substantial (30 cells); • moderate (24 cells); ∘ vacant (6 cells; no MR exercised).

Each PUT averages 24.3 mutants in the cross-source pool (range 10-30), for 292 unique v4 mutants across the 12 PUTs; each mutant is evaluated in its relevant cells over the 60-cell design with N=20 AVP repetitions per cell, and K_eq = 1000 input samples for E2 evaluation. Detailed pool counts and per-class operator specialisations are in **Appendix B.2**.

The cell density notation distinguishes ••substantial (30 cells; aligned slice + strong off-diagonal cells where MR coverage is dense and mutant detection is expected), •moderate (24 cells; cross slice with non-trivial expected detection rate), and ∘ vacant (6 cells; historically empty cells where no MR is exercised). The 12 aligned ($j = k$) cells are precisely the on-diagonal cells; the 48 cross ($j \neq k$) cells partition into the 24 •moderate + 18 ••off-diagonal substantive + 6 ∘ vacant cells. The •moderate / ••substantive distinction does not change SMS computation; it tracks expected MR coverage density per the infrastructure documentation and informs the §5.2 R_sem / R_kill decoupling discussion.

Per-class operator specialisations (illustrative). Each meta-operator (mut_C / M / G / T / F = CE / OS / HP / TF / SI in their dual form) requires PUT-class-specific specialisation:

- **mut_C Conservation-breaking:** A1 Lorenz adds $\varepsilon_{\text{drift}}$ to RHS (slow Hamiltonian drift); A2 LU decomposition omits the (k+1)-th row multiplier; B1 Beta-Bin posterior omits normalisation; C1 GPR covariance omits positive-definite diagonal term; D1 MLP backprop omits one gradient term.
- **mut_M Monotonicity-breaking:** A3 FDM Δt occasionally negative; B2 MCMC acceptance min(1, r) → min(0.95, r); C2 PCE high-order coefficient sort inserts inversion; D2 SVM decision-function sign flips near boundary.
- **mut_G Convergence-breaking:** A1 Lorenz RK4 → 1.5-order hybrid; A3 FDM 2nd-order difference → 1st-order; B3 MC doubling sample size does not 1/N-reduce variance; C3 NN-Surr training-epoch truncation.
- **mut_T Trajectory-distorting:** A1 Lorenz state-vector y / z swap; B2 MCMC inserts independent-sampling segment; C3 NN-Surr training-target slow phase shift; D1 MLP hidden-layer periodic-mask activation.
- **mut_F Fidelity-order-breaking:** A2 LU partial pivoting degrades to no pivoting; C1 GPR length-scale switches to coarse prior; C2 PCE high-order term randomly retains low-order; D3 LR regularisation occasionally large.

The per-class HP / OS / TF substitution rules give similarly differentiated specialisations across PUT classes (e.g., HP on class a = tolerance / max_iter, on class c = GPR `noise_level` / `length_scale`, on class d = MLP `hidden_dim` / dropout; OS on class a = numerical-linalg API swap `det` ↔ `sum(diag)`, on class b = sampling-API swap, on class c = surrogate-class swap GPR ↔ RBF ↔ NN). The full PUT-class × operator specialisation grid is in Appendix B.2.

3.4 Primary MP convention

The diagonal cells `j = k` form the H2-aligned slice, the off-diagonal cells form the cross slice, and the vacant cells ∘ are not formally adjudicated.

The c-class primary MP is held at the pre-registered choice (MP5) throughout this paper. All H1, H2, H3, and H4 verdicts are rendered under this configuration on v3 (same-source) and v4 (cross-source under an identical prompt). An earlier exploratory data-driven primary-MP shift was considered but withdrawn from the analysis after a permutation

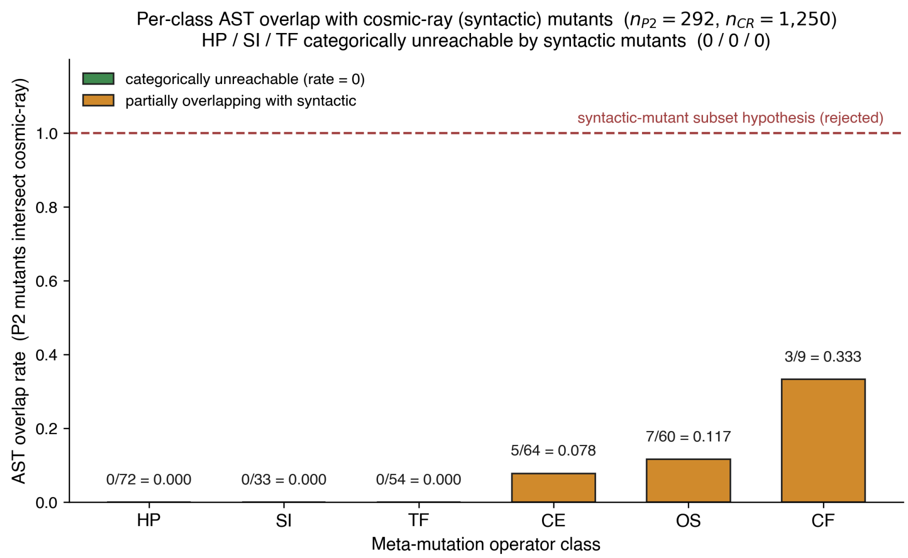

Fig. 2. Per-class AST overlap rate between 292 semantic mutants and 1,250 cosmic-ray syntactic mutants across 12 PUTs (overall 5.14%). HP, SI, TF (54.5% of the semantic pool) are categorically unreachable (0/72, 0/33, 0/54); CE 7.81%, OS 11.67%, CF 33.33% (CF is a class-specialisation, not in main 5).

null over fully exchangeable c-class (PUT, MP) cell SMS values showed the resulting δ inflation to be statistically indistinguishable from random reselection of the c-class primary MP; the corresponding pre-registration of a primary-MP-selection rule on a fresh dataset is deferred to a follow-up study.

3.5 Semantic vs syntactic mutants: 12-PUT empirical

Experimental design. For each PUT, semantic cross-source mutants (`data/mutants/${PUT}_pool_v4/`, 292 total) are compared at the AST-normalised level (`ast.dump(annotate_fields=False, include_attributes=False)`) with cosmic-ray default-operator mutants (1,250 total; `scripts/p2_vs_syntactic_ast_diff_batch.py`). Source: `data/results/cosmic_ray_12put_ast_diff.json`.

Aggregate results.

Table 7. Semantic vs syntactic mutants: 12-PUT empirical.

Metric	Value
Total semantic mutants (12 PUTs)	292
Total cosmic-ray syntactic mutants (12 PUTs)	1,250
AST-normalised overlap	15
Overall overlap rate	**5.14%**

Per-operator-class breakdown (12-PUT aggregate).

Table 8. Semantic vs syntactic mutants: 12-PUT empirical.

Class	n_p2	n_overlap	Rate	Interpretation
HP	72	0	**0.000**	Structurally unreachable
SI	33	0	**0.000**	Structurally unreachable
TF	54	0	**0.000**	Structurally unreachable
CE	64	5	0.078	Boundary class (§3.2 partial (b))
OS	60	7	0.117	Partial incidental hits
CF	9	3	0.333	b2 only, n=9

Interpretation: refuting the "post-classification copy" challenge.

- HP, SI, and TF (159 of 292 = 54.5% of v4 mutants) are **unrepresentable under default first-order configurations** of cosmic-ray's AST-local operators such as BinOp, Compare, and NumberReplacer. The unreachability is structural under first-order operators; whether HOM-based compositions [28] (not refuted) can close the gap is residual threat R12, addressed conceptually in §3.6(b).
- The CE class shows 7.81% incidental overlap, mostly LLM-generated half-step perturbations on a2 and b3 such as `_RHO=28.0` → `27.5`. The remaining 92.19% of CE mutants are AST-disjoint, because the LLMs prefer domain-aware perturbations over integer ±1 changes.
- An earlier draft labelled the OS row "× tool-inexpressible". The 12-PUT data refines this categorical claim to "△ 88.33% disjoint, 11.67% incidental hits". The systematic-versus-incidental argument in §3.6 (with details in Appendix B.1.5) clarifies why those incidental hits are stochastic byproducts rather than systematic semantic mutation.
- 94.86% of the v4 mutants cannot be reproduced by cosmic-ray defaults. The two pools occupy systematically distinct mutant spaces.

A multi-tool cross-comparison was not run because mutpy is incompatible with Python 3.10+, and mutmut's operator set overlaps strongly with cosmic-ray's. The DeepSeek, Claude, and GPT contributions to the 15 overlap files are uneven (DeepSeek 11, Claude 4, GPT 0). Appendix B.3 covers the cosmic-ray a1 single-PUT pre-12-PUT pilot, and Appendix F discusses the LLM-source distributional-shift threat R8.

3.6 Preventive-defence framing

Scope of the claim. The preventive-defence claim below is conditional on a first-order syntactic baseline; HOM-based syntactic compositions are an open question (R12) and are not refuted by the §3.5 evidence.

The HP, SI, and TF zero-overlap result, combined with the §3.2 necessary-conditions argument, amounts to a *preventive-defence* argument: semantic mutation operators address a class of fault hypotheses that lie beyond the reach of default first-order syntactic configurations (HOM not refuted; see §3.6(b)). Three points sharpen this framing.

(a) Systematic versus incidental. Satisfying conditions (a), (b), or (c) is sufficient for a single semantic mutation, but only when satisfaction reflects design intent, not stochastic byproduct, does it constitute a *systematic* semantic mutation method. A syntactic tool that occasionally hits (a) or (c) with its 12 default operators does so at a non-zero probability, but the hits are not repeatable and they carry neither of the two engineering goods we want. First, designing a semantic mutator like OS `det` → `sum(diag)` requires knowing that these expressions are equivalent on diagonal matrices but not on general matrices, a deepening of source-code understanding the syntactic tool does not exhibit. Second, domain-semantic faults, wrong physical constants, unit conversions, boundary conditions, hyperparameter semantics, numerical-method order, are not AST-local; the syntactic-tool design goals (operator typos, off-by-one errors, negation flips) do not target them. Systematic semantic mutation therefore requires (a), (b), or (c) to be design intent. Appendix B.1.5 records this argument in full.

(b) HOM caveat. HOM [29, 32] could in principle compose syntactic mutants, for example, an Arithmetic Operator Replacement combined with a Statement Deletion, to partially simulate the effects of OS or HP on some PUTs. The §4 empirics did not run a HOM comparison; we list HOM equivalence testing as residual threat R12 (Appendix F.2) and confine our "tool unreachability" claim to *first-order* syntactic tools (mutmut and cosmic-ray default configurations).

Conceptual analysis of HOM reachability. Higher-order mutation (HOM; 28, 29, 32) composes multiple first-order operators on the same source location. A natural challenge to this paper's preventive-defence framing is whether HOM compositions of cosmic-ray's 13 default operators or mutmut's 14 default operators can reach the §3.2 necessary conditions (a) cross-function-boundary, (b) domain-knowledge-dependent, or (c) algorithmic-class-change. We address this conceptually without empirical HOM testing.

For (a): HOM compositions remain over first-order operators that are AST-local within their target node; chaining AOR ∘ Statement-Deletion within a single function does not introduce a cross-module substitution like `det(M)` → `sum(np.diag(M))`, which presupposes that `np.diag` is in scope and semantically equivalent on diagonal matrices but not on general matrices. HOM can reach a kindred destination only if both operators happen to coincide with a meaningful cross-boundary swap, which is exponentially unlikely under the default operator menus.

For (b): HOM has no awareness of which constants are load-bearing in the PUT class. A two-operator chain such as Number-Replacer (1e-4 → 1e-3) ∘ Boolean-Swap on a Gaussian-process kernel does not "know" that 1e-4 is the regularization knob; it modifies the literal numerically without crossing the domain-semantics boundary. The Hyperparameter operator class (HP, mut_G) is by construction parameter-aware and class-specific, a property HOM compositions do not exhibit by default.

For (c): Algorithmic-class change (e.g. polynomial → spline basis; Lorenz RK4 → 1.5-order hybrid) requires structural code rewrites at the level of function bodies, library imports, or iteration schemes. HOM compositions over default first-order operators are bounded above by AST-local edits and so cannot reach algorithmic-class change without being effectively equivalent to a manual rewrite.

We therefore assert that the §3.5 0/0/0 unreachability for HP / SI / TF, while strictly empirical only for first-order configurations, is unlikely to be refuted by default-operator HOM compositions on combinatorial grounds. A direct empirical HOM falsification (generating mutmut second-order mutants and re-running the AST overlap analysis) remains residual threat R12.

A multi-syntactic-tool cross-comparison (mutmut and mutpy) was not run: mutpy is incompatible with Python 3.10+, and mutmut's operator set strongly overlaps cosmic-ray's. The full operator-class cross-reference between cosmic-ray (13 default operators) and mutmut (14 default operators), including which §3.2 necessary conditions each operator family reaches, is in Appendix B.6. The aggregate of the two default sets (~21 distinct first-order AST-local operator classes) collectively fails all three §3.2 necessary conditions.

(c) Source distributional shift. The 15 cosmic-ray AST overlaps are not evenly distributed across the three LLMs (DeepSeek 11/15, Claude 4/15, GPT 0/15; see `cosmic_ray_12put_ast_diff.json`). DeepSeek tends to generate syntactically simpler mutations. This LLM-source bias is discussed as R8 (Appendix F.2) but does **not** affect the systematic-vs-incidental argument, which is based on *categorical* AST-locality, not hit frequency: the HP / SI / TF zero-overlap result is 0/0/0 across all three sources.

3.7 Adjoint extension arm: subjects

The adjoint invariant ψ_6 (§2.7) is exercised on a separate, confirmatory set of three third-party linear-operator libraries: scipy `LinearOperator` (linear algebra), pylops real-FFT (signal processing), and jax `linear_transpose` (automatic differentiation), disjoint from the 12 PUTs and the pre-registered 60-cell. Each library carries a real fixed defect in its adjoint code path as a ground-truth anchor (scipy #8900, pylops #292, jax #6223). For each, the MR families are adj.self (self-adjointness) and adj.dual (scaled/dual-operator transpose), and the mutant pool mixes adjoint-breaking mutants (the real defect among them) with semantically equivalent mutants a strong MR must *not* kill; the real defects are reproduced in extracted-diff form because the pre-fix releases are uninstallable on the host (no `arm64` wheel). The arm is confirmatory and additive: it does not alter the 60-cell denominator, the primary-MP convention, or the pre-registered hypotheses. Full subject specifications and fault mechanisms are in supplementary Appendix H; results in §4.9.

3.8 Procedure overview

With the subjects, operators, and the adjoint extension arm fixed (§3.1–§3.7), we now specify the procedure that produces each cell's measurements.

Pipeline order: §3.1 PUTs → §3.9 Mutant generation → LRCA L0 prescreen → §3.11 Equivalence ($E_1 \wedge E_2$) → AVP → killed/survive → LRCA three-layer diagnosis → SMS + C_1_share report.

Each (i, k, j) cell is reported with `inst_count`, `equiv_count`, `killed_count`, `survive_count`, SMS_{i,k,j}, C1_share_{i,k,j}, suspect_share_{i,k,j}, and the three rates `inst_rate / equiv_rate / survive_rate` corresponding to RQ3.

3.9 Cross-source protocol

To isolate the contributions of LLM same-source bias and MR-MP alignment design to Cliff's delta, we ran a three-stage ablation:

Table 9. Cross-source protocol.

Version	Mutant pool	c-class primary MP	Use
v3	Same-source (Claude Opus 4.6)	MP5 (pre-registered)	H2 baseline (pre-registered primary)
v4	**Cross-source** (Claude Opus 4.6 + GPT-5.4 + DeepSeek chat)	MP5 (held at pre-registered)	Isolate LLM-source diversity

The v4 protocol (`scripts/cross_source_campaign.py`) runs each (PUT, operator) pair on Claude / GPT / DeepSeek with K=3 trials under an identical prompt template (temperature 0.7, V1-V4 mechanical-validation gate). Cross-source pool capacity: 37 operators × 3 sources × 3 trials = 333 attempts, 89.5% V1-V4 pass, 298 confirmed mutants distributed nearly equally (Claude 101, GPT 98, DeepSeek 99); final-stage rejection of duplicate and QA-failing mutants leaves the 292 that enter the v4 pool used for SMS and the AST audit.

Declared confound: protocol asymmetry (R13). v3 used the original Phase-1 dual-blind reviewer protocol (Claude generation + GPT-5.4 review + DeepSeek arbitration); v4 passes V1-V4 mechanical gates only and does **not** invoke a reviewer LLM. A fraction of the v3 → v4 source-axis contrast may therefore reflect a slight quality shift rather than LLM-source diversity per se. The follow-up study will rerun dual-blind on the v4 grid; full per-LLM token / latency / cost figures, V1-V4 specifications, and the full differential-prompt protocol are in **Appendix C.1**.

3.10 Mutant-pool prescreen and equivalence detection

Pool prescreen (LRCA L0). Each candidate mutant passes three gates:

(1) Static lint and type checking.
(2) A unit self-test on simple inputs to confirm that the PUT loads, runs, and returns a finite output.
(3) A double-blind review sign-off under the Phase-1 protocol.

In Phase-1, the generator LLM is Claude Opus and the reviewer LLM is GPT-5.4; the two roles are isolated, and the reviewer sees only the PUT and the mutant code and outputs a (syntactic, executable, fault-injected) triple. Inconsistent reviewer outputs go to a manual arbitration queue (no more than 10% of cases); double-confirmed mutants enter the pool. We then randomly sample 20% of the pool for manual review by scientific-computing researchers, and downgrade the entire batch to manual review if manual–reviewer inconsistency exceeds 10%. Each cell produces 30–50 candidates and retains 10–15 mutants. The v4 cross-source pool does **not** invoke the reviewer LLM, for cost and speed reasons; a fraction of the v3 → v4 source-axis shift may therefore reflect a quality decline rather than a source-diversity contribution. We declare this protocol asymmetry as threat R13 (§5.1; full justification in Appendices C.2 and C.3).

Equivalence detection (E1 ∧ E2). For each mutant `s'` in mut_j(S_i):

(a) **E2 step.** Sample K_eq = 1000 inputs `x ~ D_S`; compute $\|S_i(x) - s'(x)\|$; if $\leq \varepsilon_{\text{eq}}$ for all samples, mark E2-pass.
(b) **E1 step.** For every `mr ∈ MR_{i,k}`, compare `AVP(S_i, mr)` with `AVP(s', mr)`; if all consistent, mark E1-pass.
(c) **Conjunction.** E1 ∧ E2 → enter `equiv_{i,k,j}`, exclude from SMS denominator. Otherwise pass to AVP-killed determination.

E2 is run before E1 because E2 is computationally cheaper (K_eq scalar evaluations) and quickly discards numerically distinct mutants; the remaining candidates are screened by AVP for MR-relation equivalence. The joint condition is conservative: false-non-equivalent (E1 ∨ E2 fail) is much easier than false-equivalent (both must mis-pass simultaneously), biasing SMS slightly *high*, an explicit conservative engineering choice (Appendix A.3).

Killed determination. With OR-aggregation across mr in MR_{i,k}:

$$\begin{aligned}\text{killed}(s', \text{MR}_{i,k}) \iff \exists\, mr \in \text{MR}_{i,k} :\\ \text{AVP}(S_i, mr) = \text{pass} \wedge \text{AVP}(s', mr) = \text{fail}.\end{aligned}$$

Each (i, k, j) cell is sampled N = 20 times (statistical replicate) to compute the per-cell SMS, with mutant-level fail ratios feeding the LRCA L1 decision. AVP version is pinned to the upstream infrastructure commit hash (`<AVP-vX.Y>`);

our reproducibility package embeds the complete AVP source so that the package remains self-consistent under any upstream evolution.

3.11 LRCA three-layer diagnosis

For each killed mutant the three-layer decision tree applies in order:

- **L1 tolerance robustness** (all classes). N=20 replicates with a fail-ratio cutoff at 0.80; a kill that survives less than 80% of the 20 replicates is labelled C2 (numerical-tolerance perturbation) and exits.
- **L2 OOD triage** (C / D classes only). Sample from `D_S^valid`; if the mutant fails *only* in the OOD region, label C3 (out-of-distribution) and exit.
- **L3 statistical-assumption baseline** (B / D + Wilcoxon / DTW only). Pre-check IID / stationarity on the PUT's own repeated samples; if the AVP statistical assumption is itself violated, label C4 and exit.
- **Artefact recheck**. An external reviewer re-examines mutant code + prompt history; if LLM / artefact evidence (e.g., mutator over-injection) is found, label C5; otherwise label C1.

Multi-label priority C5 > C4 > C3 > C2 > C1 takes the earliest confirmed non-semantic cause (decision-tree control flow). Threshold calibration over a 9-grid (`ood_band ∈ {0.02, 0.05, 0.10}` × `tolerance_multiplier ∈ {3.0, 10.0, 30.0}`, repeats fixed at 20) lifts H4 from 10/60 to 12/60 cells; the best combination (`ood_band = 0.02, tolerance_multiplier = 3.0`) is reported as primary, with default-threshold results retained as control (`lrca_60cell_v3.json`). The calibration ceiling (12/60 = 20%) remains far below the 80% pre-registered threshold; H4 is unattainable on this dataset as an intrinsic property of LLM-mutant pools, not a calibration issue. Detailed grid table, L0-L3 sub-protocols, and decision-tree pseudocode are in **Appendix A.2 + C.4**.

4 Results

4.1 Statistical pipeline

The primary statistical reporting follows a pre-registered hierarchy:

Table 10. Statistical pipeline.

RQ	Primary statistics	Reporting format
RQ3	inst_rate, equiv_rate, C1_share, survive_rate	60-cell heatmap + 4-class marginals
RQ4	aligned-SMS vs cross-SMS; sparse ∘ vs dense ∘∘equiv_rate	Cliff's delta + odds ratio + 95% bootstrap CI
RQ5	ΔSMS_c ($c \in$ {A,B,C,D}); CV(ΔSMS)	sign test (df = 3) + descriptive forest plot
RQ5 (desc.)	Spearman ρ + Kendall τ (SMS vs PC)	scatter + dual-metric ranking comparison

This pipeline answers the empirical questions RQ3–RQ5; RQ1 is answered by Theorem 9.1 (§2.6) and RQ2 by the AST-normalised structural audit (§3.5), outside the statistical pipeline.

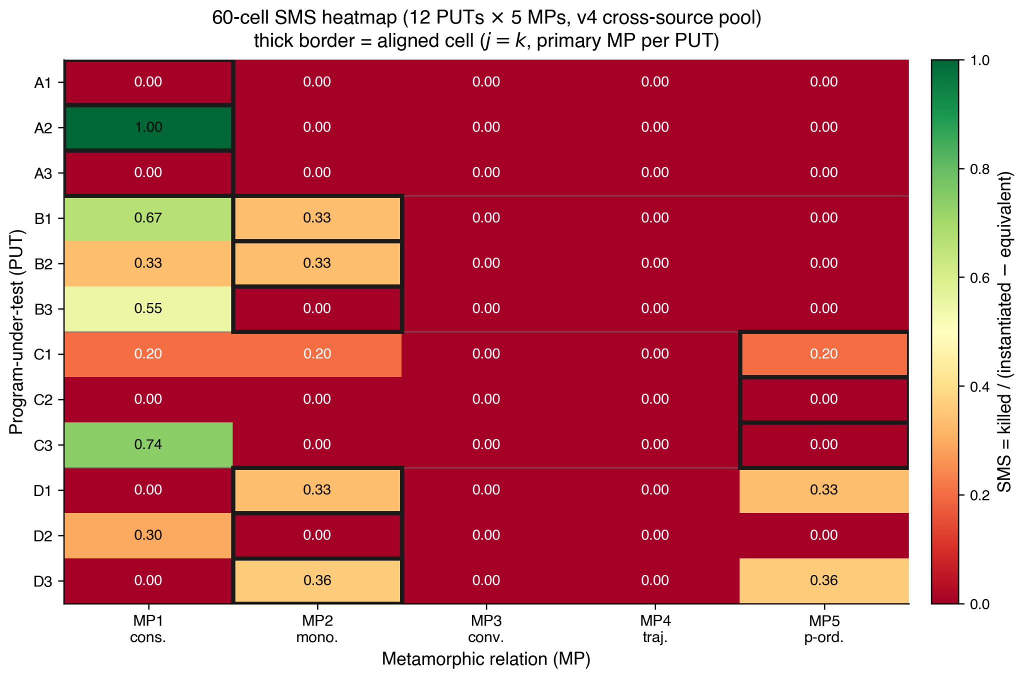

Fig. 3. 60-cell heatmap of mean SMS over 12 PUTs × 5 MPs (v4 cross-source primary). Diagonal cells (operator-MP aligned) show concentrated mass; off-diagonal mass reflects bleed into adjacent MPs. Empty rows = PUT-level zero-mass cohort (PUTs A1, B1, D2; see §5.2).

Where multiple per-class tests are reported (the per-class Friedman tests of §4.5), they carry a Bonferroni correction across the four classes; we do not report a separate cell-level family of p-values, so no family-wise cell-level correction is claimed. Bootstrap 95% CIs use 1,000 iterations as default (B = 10,000 for the headline H2 delta CI). Mixed-effects modelling (`sms ~ C(class) * C(operator) + (1 | put)`) failed with a Singular matrix at N = 60: the class × operator interaction design matrix is rank-deficient and the PUT random-intercept variance collapses to the boundary at this sample size. The Friedman test is the non-parametric formal alternative. The four hypotheses are pre-registered: H1 (operator implementability) requires $\geq$ 4 of 5 operators producing $\geq$ 5 non-equivalent mutants on $\geq$ 9/12 PUTs; H2 (aligned-vs-cross) requires odds ratio $\geq$ 3.0 *and* Cliff's delta $\geq$ 0.474 (Romano 2006 large-effect threshold); H3 (cross-class consistency) requires sign-test 4/4 across 4 classes plus CV(ΔSMS) < 0.5; H4 (LRCA mass) requires mean suspect_share $\leq$ 0.20. SMS variants (SMS, SMS_unfiltered) and construction lemmas are in **Appendix D.1**.

4.2 RQ3: 60-cell distribution

Table 11. RQ3: 60-cell distribution.

Metric	Value
Number of cells	60
Mean SMS	0.104

Table 11. RQ3: 60-cell distribution.

Metric	Value
Median SMS	0.000
Std SMS	0.213
Cells with SMS = 0	45 / 60
Mean mutants / cell (v4)	24.3 (range 10-30)

The **zero-mass dominance** (45/60 = 75%) is concentrated in the cross-MP slice: ~88% (42/48) of cross cells are zero, ~25% (3/12) of aligned cells are zero. Cliff's delta is well-defined under this distribution but its inference is effectively dominated by `n_aligned = 12 + n_cross_nonzero ≈ 6 = 18`, not the surface n=60. Median odds ratio is formally infinite (median(cross) = 0); we report "aligned median > 0 = cross median" as auxiliary qualitative evidence.

Effective-n note. The surface n_aligned = 12 and n_cross = 48 mask a much smaller information content: only about 18 observations carry nonzero SMS (12 aligned + ≈ 6 nonzero cross). This is a heuristic information count, not a formal effective sample size, but it explains the wide 95% bootstrap CI [0.127, 0.740] (upper/lower ratio ≈ 5.83, consistent with the known liberal tendency of the percentile bootstrap when few observations carry signal). The implications for power and the H2 verdict direction are quantified jointly with the stipulated-alternative analysis in §4.4. PUT-class diversification at $n \geq 30$ is a testable route to relaxing the effective-n constraint. Power and effect-size disambiguation are treated jointly with the stipulated-alternative analysis in §4.4 (avoiding repetition).

Consistency with LLM-mutant literature. Tip et al. [37] LLMorpheus reports high cross-MP failure proportions on JavaScript scientific-computing PUTs (specific numbers not listed in cited literature); zero-mass dominance appears to be a shared characteristic of LLM-mutant + existing MR-MP alignment designs, not a quirk of this paper's PUT selection.

H1 verdict: not met. H1 (RQ3) requires at least 4 of the 5 operators to produce $\geq$ 5 non-equivalent (confirmed) mutants on at least 9 of the 12 PUTs. Table 12 counts confirmed non-equivalent mutants per operator class across the 12 PUTs: only the Hyperparameter operator clears the $\geq 9/12$ bar.

Table 12. H1: PUTs reaching $\geq$ 5 non-equivalent mutants, per operator class.

Operator class	PUTs with $\geq$ 5 non-equiv mutants (of 12)
Conservation Erosion (CE)	4 / 12
Operator Substitution (OS)	5 / 12
Hyperparameter (HP)	**9 / 12**
Trajectory Flip (TF)	5 / 12
Structural Injection (SI)	1 / 12

Because the semantic operators are class-targeted, each is applicable to a subset of the four PUT classes rather than uniformly to all twelve , only the broadly-applicable Hyperparameter operator clears the 9-PUT threshold. Thus 1 of 5 operators meets the criterion and **H1 is not met as pre-registered**; the uniform-applicability assumption behind the $\geq 9/12$ bar does not hold for class-targeted operators. The per-operator implementability profile (r_{sem}, d_{impl}) is reported in Appendix C.4.

4.3 RQ4: Aligned vs cross

Table 13. RQ4: Aligned vs cross.

Slice	n	Mean SMS	Median SMS
aligned (j = k)	12	0.275	0.267
cross (j ≠ k)	48	0.061	0.000

Two delta point-estimates under the pre-registered primary-MP convention:

- **v3 (same-source, pre-registered):** delta = **0.323**, 95% CI [0.017, 0.622]
- **v4 (cross-source, c-class held at MP5):** delta = **0.314**, 95% CI [0.014, 0.622]. The cross-source pool aggregates Claude / GPT / DeepSeek under an identical prompt; the c-class primary is held at the pre-registered MP5 so that the v3 → v4 contrast isolates the LLM-source-diversity axis without inheriting any post-hoc primary-MP selection.

H2 verdict: not met. H2 carries two pre-registered criteria, Cliff's $\delta \geq 0.474$ and an aligned-to-cross odds ratio ≥ 3.0. Neither delta crosses the Romano et al. [36] large-effect threshold 0.474, 0.323 and 0.314 both fall just below the Romano et al. [36] medium threshold of 0.330, so the delta criterion fails. The odds ratio is degenerate: because the cross-slice median SMS is exactly 0 (table above), the median odds ratio is $+\infty$, so the ratio criterion is satisfied only trivially under zero-inflation and carries no evidential weight. H2 is therefore not met as a large-effect target. The contrast $\delta_{v4} - \delta_{v3} = -0.009$ (95% CI covers zero) supports a null reading on the LLM-source-diversity axis under an identical prompt: replacing a same-source pool with a three-LLM cross-source pool does not appreciably shift Cliff's delta within this design. We frame H2 as an underpowered exploratory contribution and defer full verification to a follow-up study with a larger PUT sample.

This medium-effect placement is consistent with Tip et al. [37] LLMorpheus's medium-effect range on JavaScript LLM mutants, a contextual literature observation (estimand caveat: their delta compares "LLM vs traditional mutants on fault detection", ours compares "aligned vs cross MP slice within one pool"; the numerical similarity is not substantive support).

4.4 RQ4: Stipulated-alternative power

The §4.2 effective-n constraint motivates an explicit power analysis. A plug-in bootstrap (5,000 replications, seed = 42) samples with replacement from the observed (n = 12, n = 48) v4 SMS distributions. The bootstrap exceedance-probability table (probability that the resampled $\hat{\delta}$ exceeds each threshold under the observed distribution, not power against a stipulated alternative) is:

Table 14. RQ4: Bootstrap exceedance probability under the observed distribution.

Threshold	Interpretation	Exceedance prob. at (12, 48)
$\delta > 0.000$	Any effect	**0.997**
$\delta > 0.147$	Small	0.966
$\delta > 0.330$	Medium	0.759
$\delta >$ **0.474**	**Large (H2)**	**0.423**

The plug-in result answers the question "given the *observed* distribution, how often do we exceed the threshold?" A stipulated-alternative simulation answers a more pointed question: "if the *truth* equals the H2 boundary 0.474, how often does the (12, 48) design return $\delta \geq 0.474$?" SMS distributions have heavy ties at zero (45 of 60 cells), so a raw shift is discontinuous: any $\varepsilon > 0$ jumps δ from 0.314 to 0.74. We therefore use a mixture, drawing the aligned sample with probability w from (observed_aligned + 0.001) and with probability 1 − w from observed_aligned, calibrating w = 0.094 to realise $E[\delta] = 0.4746$.

Table 15. RQ4: Stipulated-alternative power.

Stipulated truth	Criterion	Power
0.474	$\hat{\delta} \geq 0.474$ (point estimate)	**0.491**
0.474	Lower 95% CI bound > 0 (any effect)	0.868

Even when the truth equals the H2 boundary, this design returns "not met" verdicts in roughly half of replications. This supports the framing in §4.3: the H2 verdict is a factual statement about the point estimate failing to clear the threshold, not a claim that the effect is necessarily smaller than 0.474. Increasing the sample size narrows the confidence interval but cannot lift the point estimate. The plug-in sample-size sweep (n_aligned ∈ {6, 12, ..., 60}; n_cross = 4 × n_aligned; power for $\delta > 0$ reaches 0.974 at n_aligned = 6 and 0.996 at 12, then plateaus) is in Appendix D.3.

The source-axis contrast is underpowered too. We did not run a separate power simulation for the source-axis contrast $\Delta\delta = -0.009$ (CI covers zero); its estimand and dependence structure differ from the aligned-vs-cross δ, so the 49.1% figure above does not transfer to it. The small design (n = 12 PUTs) leaves the −0.009 null consistent with a wide range of true source-diversity effects, so we explicitly do not read it as evidence that source diversity is inert. A properly-powered strong-sense test is deferred to a follow-up study.

Note on monotone-transformation invariance. Cliff's δ is rank-based: a function of $U/(n_1 \cdot n_2)$, so applying a logit transform to SMS gives $\delta_{\text{logit}} \equiv \delta_{\text{raw}}$ by construction. This is consistent with the rank-invariance theorem and does **not** constitute additional robustness evidence. The genuine robustness threats to the H2 verdict come from the zero-mass dominance (§4.2), not from any metric-scale choice.

4.5 RQ5: Cross-class consistency and Friedman

Table 16. RQ5: Cross-class consistency and Friedman.

Class	Mean SMS (v3)	Mean SMS (v4)
a (numeric)	0.067	0.067
b (probabilistic)	0.156	0.148
c (surrogate)	0.047	**0.089 (+91.5%)**
d (ML)	0.081	0.112 (+38.3%)

Sign test (within-class aligned mean − cross mean, sign = +): **3 / 4 (partial) under both v3 (same-source) and v4 (cross-source)**; the b-class is the inverted cell.

H3 verdict: not met. H3 carries two pre-registered criteria, a within-class sign test of 4/4 and CV(ΔSMS) < 0.5. The sign test is 3/4 (partial) under both v3 and v4, cross-source pooling does not flip the b-class inversion, so the consistency criterion fails. The per-class effects ΔSMS_c (aligned mean − cross mean) span from negative in the inverted class to large positive elsewhere, giving CV(ΔSMS) well above 1, far exceeding the 0.5 dispersion threshold, so the dispersion criterion also fails.

The mixed-effects primary model `sms ~ C(class) * C(operator) + (1 | put)` returned Singular matrix; the fallback `sms ~ C(class) + C(operator) + (1 | put)` had PUT random-intercept variance hit boundary 0 (degenerate). We therefore use Friedman as the non-parametric formal alternative:

- **PUT (n=12) × MP (k=5): chi^2 = 15.30, p = 0.0041** (significant).
- MP rank means: 2.92, 2.58, 2.08, 3.08, 4.33.
- Per-class Friedman with **Bonferroni × 4 correction**: a 1.000 / b **0.116** / c 1.000 / d 1.000, no per-class result remains significant after correction. Kendall's W effect sizes: a 0.333 (small) / b 0.898 (large concordance, but caveat: N=3 per class makes this label nominal only) / c 0.333 / d 0.417.

Caveat. Friedman tests "are MP rank differences present" (averaged over PUTs); H3 tests "is direction consistent across 4 classes". These are logically independent. H3 stands on the §4.5 sign test; per-class Friedman is descriptive only (small N=3 per class).

4.6 RQ5 (descriptive): SMS vs Pattern Coverage

Status. The SMS–Pattern-Coverage relationship, a descriptive sub-question of RQ5, is reported as a descriptive observation, not a hypothesis test, because n = 12 places the 95% Spearman CI at roughly [−0.5, +0.6], so the test cannot distinguish zero, moderate-positive, or moderate-negative correlation. The numbers below are recorded so that a future study (n ≥ 30 PUTs) can pre-register a directional hypothesis.

Pattern Coverage (PC) per PUT = #triggered (MP_k, R_outcome) cells / 10. Range [0.500, 1.000], mean 0.733. Pairing with mean SMS over 5 MPs: Spearman rho = **0.163** (p = 0.613); Kendall tau = 0.136 (p = 0.568); n = 12.

Statistical-power caveat first. At n = 12, Spearman's 95% CI is approximately [-0.5, +0.6] (Fisher-z approximation, a rough small-sample heuristic); the test cannot distinguish "zero correlation" from "moderate positive correlation" or "moderate negative correlation". p = 0.61 / 0.57 does **not** support the strong claim that "SMS is independent of PC" or the strong claim that "SMS is strongly correlated with PC". The conservative finding is *no detectable correlation at n = 12*; orthogonality is a hypothesis for a future study (n ≥ 30 PUTs or refined PC operationalisation; full PC operationalisation in Appendix D.5).

4.7 H4 (RQ4): LRCA mass

Table 17. H4: LRCA mass.

Metric	Value
Mean C1_share (default threshold)	0.164
Mean C1_share (calibrated best, ood_band = 0.02)	**0.209**
Mean suspect_share (calibrated best)	**0.791**
Cells with suspect_share ≤ 0.20 (secondary)	12 / 60 = 20.0%

H4 verdict: not met. The pre-registered criterion is mean suspect_share ≤ 0.20; the observed mean is **0.791** (v4), decisively above the threshold, so H4 fails on its own metric. A dense cutoff sweep (Appendix D.2) confirms this is an intrinsic property of LLM-mutant pools rather than a calibration artefact: the secondary per-cell pass-ratio (cells with suspect_share ≤ 0.20) is flat at 12 / 60 = 20% across cutoffs $\in [0.05, 0.40]$, because v4's suspect_share distribution is severely bimodal (median 1.0, with 48 cross cells near 1 and 12 aligned cells near 0; the [0.20, 0.80] interior is nearly empty). H4 was pre-registered before pool characteristics were known, so this is a finding worth reporting.

4.8 RQ4: The strong boundary, false positives and false negatives

The 60-cell audit (§4.2–§4.7) exercises strong MRs in the regime where Theorem 2 holds. To map the strong boundary of Proposition 2 we run a complementary study on six third-party or textbook programs spanning the meta-patterns, kept separate from the 12 PUTs. Wherever a genuine fixed defect exists in a third-party library we use real version-switching: the pre-fix and post-fix releases run unmodified in isolated environments, and we record the repository, the version pair, and the issue/PR, and we fall back to a textbook faithful implementation only when the real defect cannot be exercised on the host (no `arm64` wheel, a heavyweight Fortran build, or a drift below machine resolution). ε_{tol} is an explicit factor in every case.

Table 18. RQ4: Strong-boundary map over six programs.

Family	Program (provenance; method)	Strong MR (ε_{tol})	Correct / faulty verdict	Boundary role
inv.con	non-conservative Burgers; [24] (textbook)	shock speed (0.03)	0.498 pass / −0.002 killed	strong
inv.eqv	scipy `align_vectors` 1.13.0 → 1.13.1, #20660 (real)	rotation residual (10^{-6})	10^{-16} pass / 2.0 killed	strong
adj.self	scipy `_adjoint` #8900/PR#8962 (real, extracted-diff)	adjoint-identity residual	10^{-16} pass / 1.4–2.0 killed	strong
rev.traj	leapfrog vs explicit Euler; [23] (textbook)	energy-drift ratio (≤ 3)	1.00 pass / 1952 killed	strong

PINN	soft vs hard BC; DeepXDE #26 (real torch)	exact BC $\lvert u(0)\rvert$ (10^{-4})	hard 0 pass / soft 5×10^{-4} killed	FP (weak MR)
struct. (FN)	numpy float32 RNG 1.21.6 → 1.22.0, #17478 (real)	mean / range / dist	both pass; buggy survives	FN (surviving)

Strong regime (four cases). For inv.con, inv.eqv, adj.self, and rev.traj the correct program satisfies the strong MR within ε_{tol} while the faulty program is killed, exactly as Theorem 2 predicts. Two are real fixed library defects (scipy `align_vectors`; scipy `_ScaledLinearOperator._adjoint`, whose pre-fix release has no `arm64` wheel and is therefore exercised by applying the exact fix diff in reverse); two are textbook contrasts standing in for a real defect that cannot be exercised on the host, a conservative versus non-conservative Burgers scheme for the GeoClaw `filpatch` conservation bug, and a symplectic leapfrog versus explicit Euler integrator for the REBOUND IAS15 energy drift, whose real signature near 10^{-14} lies below resolution.

False positive, weak MR (PINN). A physics-informed network that enforces the boundary condition as a soft penalty is a legitimately trained, non-faulty model, yet its exact-boundary residual $\lvert u(0)\rvert \approx 5\times 10^{-4}$ exceeds a strict $\varepsilon_{\text{tol}} = 10^{-4}$, so the exact-boundary strong MR flags it: the discretisation (the soft constraint), not a fault, breaks the invariant. This is the discrete-program instance of the rotational-symmetry counterexample, a continuous invariant reproduced only approximately, and the empirical face of Proposition 2(i). The hard-constraint ansatz $u = x(1-x)\,\mathrm{NN}(x)$ satisfies the invariant by construction and passes, confirming that the violation is a property of the discretisation, not of training.

False negative, surviving non-equivalent mutant (RNG). The numpy float32 generator defect (#17478) zeroes the lowest mantissa bit of every sample; the structural MR family, mean $\approx$ 0.5, range $[0,1)$, distribution shape, is satisfied by both releases within ε_{tol}, so the genuinely non-equivalent buggy version survives. This is coincidental satisfaction of the MR, and because the fault signature lies below the structural tolerance it is a tolerance fault; the surviving mutant is non-equivalent (an independent odd-bit-fraction discriminator reads 0.0 for the buggy release versus $\approx$ 0.5 for the correct one), and therefore distinct from a classical equivalent mutant. It is the empirical face of Proposition 2(ii).

The ε_{tol} trade-off. The PINN case makes the sensitivity–specificity coupling runnable: at the strict $\varepsilon_{\text{tol}} = 10^{-4}$ the legitimately-trained soft-BC model is killed (a false positive), while at a loose $\varepsilon_{\text{tol}} = 10^{-3}$ it survives correctly; if the same residual pattern instead arose from a true boundary-condition fault, loosening ε_{tol} would turn that survival into a false negative. Tightening ε_{tol} trades false negatives for false positives and loosening it does the reverse, as Proposition 2 requires. The strong boundary is therefore the honest scope of the duality, a tunable operating point, not a defect to be removed.

4.9 RQ4: Adjoint extension arm (forward)

Where §4.8 maps the boundary at which the duality fails, a small confirmatory arm checks it holding in the forward direction for the sixth meta-pattern ψ_6. On the three third-party linear-operator libraries of §3.7 (scipy, pylops, jax), each carrying a real fixed adjoint defect, a strong adjoint MR detects only the active mutants: across all three substrates the correct program survives, every non-equivalent mutant (including the three real defects scipy #8900, pylops #292, and jax #6223) is killed, and no semantically equivalent mutant is killed, giving a forward SMS of 1.00 with zero false positives on the equivalent set (full subjects, mutant pools, and the per-substrate kill table in supplementary Appendix H). The arm is confirmatory and additive, it does not enter the 60-cell denominator and leaves H1–H4 unchanged, and carries two scope caveats: the scipy #8900 defect is shared with the §4.8 boundary study (so it is not counted as independent corroboration twice), and the pools are hand-constructed, low-cardinality sets with the real defects reproduced in extracted-diff form, so the arm demonstrates rather than measures adequacy (threat R14).

5 Discussion

5.1 Cross-source contributes mutant quality, not effect size

Going from same-source to cross-source under the pre-registered primary-MP convention shifts Cliff's δ by only -0.009 (95% CI covers zero). Cross-source pooling raises mean C1_share from 0.164 to 0.209 (a 27% relative increase), class-c mean SMS by **+91.5%**, and class-d mean SMS by 38.3%. Under an identical prompt template, three LLMs produce similar distributions for the aligned-vs-cross question. Within this fixed-prompt design, LLM source identity is therefore not the dominant factor behind the H2 ceiling: cross-source pooling improves the diagnostic quality of the kill set (LRCA C1 share, class-mean SMS) without moving the aligned-vs-cross effect size. This is a scoped null, not a general claim that LLM diversity is inert. The strong-sense source-diversity test, with per-LLM differential prompts (V_persona, V_cot), is deferred to a follow-up study. Appendix C.1 records the full protocol.

The §3.5 evidence (5.14% AST overlap, HP/SI/TF at 0/0/0) confirms that the medium-effect ceiling is not an artefact of LLM-pool overlap with the syntactic-mutant space. 94.86% of v4 mutants are AST-disjoint from cosmic-ray defaults, and the three classes unreachable under default first-order configurations (HP, SI, TF; 159 of 292 mutants; HOM not refuted, see §3.6(b)) lie outside that space by construction. A larger aligned-versus-cross effect therefore points first to MR-design refinement. A larger sample would mainly reduce uncertainty around this design unless the MR design itself changes.

A numerical-coincidence note. Petrovi'c et al.'s [2021] ~20% productive-mutant rate at Google is close to our calibrated C1_share of 0.20. Their construct is a developer survey (subjective usefulness), and LRCA's C1 is a diagnostic annotation over observed kill behaviour; the agreement between the two is contextual, not mechanism validation.

5.2 Decoupling between Rsem and Rkill

The operator-level pilot in Appendix C.4.1 reveals a sharp pattern: HP, TF, and OS operators on the c-class (surrogate) and d-class (machine learning) PUTs reach R_sem $\approx$ 1.0 (semantic feasibility, the LLM successfully produces a syntactically valid, executable, intent-bearing mutant) but R_kill = 0 under the PUT's primary MP (the AVP fails to detect the mutant under that MR). Concretely, six cells show R_sem $\geq$ 0.9 with R_kill = 0: c1_CE1 (GPR `noise_level` 1e-4 $\rightarrow$ 1e-1), c2_OS1 (polynomial $\rightarrow$ spline basis), c3_HP1 (relu $\rightarrow$ tanh), c3_TF1 (max_iter 1000 $\rightarrow$ 5), d1_HP1 (MLP α 1e-4 $\rightarrow$ 1.0), and d3_HP1 (LR C 1.0 $\rightarrow$ 1e-4). In each, the mutant *is* a valid hyperparameter-semantic injection, but the primary MP, MP5 asymptotic for the c-class and MP2 monotonicity for the d-class, is an asymptotic or statistical relation, and the parameter deviation it tolerates is wide enough to absorb the mutant.

The cell-level evidence in §4.2 reproduces this pattern: 75% of cells are zero, concentrated in the cross slices. Under v4 cross-source, mean C1_share rises from 0.164 to 0.209 (+27%): among the few killed mutants, the cross-source pool produces cleaner LRCA diagnostic labels. Source diversity therefore improves the *quality* of the kill set without expanding its *coverage*. **The engineering insight is that operator-MP alignment in MR design is necessary for strong SMS signals; merely enlarging the semantically feasible mutant pool dilutes the C1 proportion**

without lifting the kill rate. Two open questions remain: whether SMS can infer which MP class is missing for which PUT class, and whether LRCA can be calibrated with class-specific thresholds separating likely semantic failures from artefacts.

This decoupling motivates the source-axis reading in §4.3: the v3 → v4 micro-shift of −0.009 estimates LLM source diversity under a fixed prompt, while MR-design adequacy at the operator-MP level governs the kill rate in this design.

Real-defect evidence slice. A 34-case real-defect evidence slice gives a useful check on this interpretation without changing the paper into a benchmark article. We report verified result totals only: defect mining, rejected cases, repository tooling, and benchmark release are left outside the paper. The result-level ledger used here is included in the supplementary evidence package: it records the five-stratum counts, the aggregate mutation-phase statistics, the 30/30 real-defect face, and the non-nesting case identifiers, but not candidate mining or tooling. In this mutation-phase slice, the pattern-derived MR group beats literature-generic baselines under the primary all-mutant denominator (mean paired difference +0.101, Holm-adjusted $p = 0.046$, Cliff's $\delta = +0.247$), but the advantage is modest: the pattern-derived group does not significantly dominate its ablations under the same denominator. On the real-defect face, however, the distinction becomes sharper: the pattern-derived relation detects every real defect in the reported mutation-suite face (30/30), whereas generic and random relations are mostly blind, and ablations that look close in aggregate kill-rate often lose the real defect.

This pattern is consistent with the construct separation predicted by the fiber view of §2.9. A mutant kill-rate is evidence about which sampled semantic effects an MR observes; it is not, by itself, a verdict on MR quality. The same real-defect slice also records non-nesting cases in which generic baselines kill mutants that the pattern-derived relation is structurally blind to. Thus semantic mutation should be read as a certificate-bearing diagnostic for a declared semantic stratum, not as a scalar hierarchy in which one mutation family simply dominates another. This evidence therefore strengthens the paper's main argument while keeping benchmark construction outside the manuscript's contribution boundary.

5.3 H3: cross-class consistency not met (sign test 3/4 stable across sources)

All four class means are positive in v3 (same-source) and v4 (cross-source). Inter-class balance improves under cross-source (c +91.5%, d +38.3%), confirming that c / d classes have higher mutant-diversity demand than a / b. Mixed-effects unavailability (Singular) is a sample-size constraint at N = 60 / 12 PUTs, not evidence absence. **H3 verdict: not met**, the sign test is 3/4 (partial) under both v3 and v4 with the b-class inverted, and CV(ΔSMS) exceeds the 0.5 dispersion threshold. The Friedman main effect (chi^2 = 15.30, p = 0.0041) speaks to MP differentiation, not H3 direction.

5.4 Stakeholder analysis

Scope. All deployment claims in this subsection are bounded to single-output `float` → `float` kernels under 2 KB, the 12 PUTs of this paper. They do not apply to industrial-scale, multi-module, or multi-output scientific computing software, which lies beyond the scope of this paper. Table 19 gives interpretive readings for SMS and LRCA outputs; it is not an acceptance-threshold table.

Table 19. Reading SMS and LRCA together.

Observation	Interpretation	Design response
Low SMS, high C1_share	The MR family detects few mutants, but the kills it does make are mostly semantic	Add MRs for adjacent semantic strata before enlarging the mutant pool
Low SMS, high suspect_share	The MR family mainly triggers tolerance, OOD, statistical, or artefact labels	Inspect tolerances, input domain, and statistical assumptions before treating kills as adequacy evidence
High SMS, low real-defect detection	Mutant kill-rate and defect detection have diverged	Revisit the declared semantic stratum and add real-defect-facing MRs
Generic relation kills outside the pattern-derived relation	Relation families are non-nested	Treat the result as a different semantic-effect fiber, not as domination by one MR family

Test engineers can read SMS as a per-MR scalar adequacy score. When SMS sits well below the aligned baseline of about 0.275 (§4.3), the LRCA labels, C2 tolerance, C3 OOD, C4 statistical, point to specific repair paths. Air-gap incompatibility is a hard limitation: the workflow depends on external LLM API calls and is incompatible with most regulated air-gapped Verification and Validation (V&V) environments (IEC 60880, DO-178C, IEC 62304, ISO 26262). Pre-generated mutant pools can be reproduced offline, but generating new mutant pools requires LLM access. Self-hosted open-weight LLMs and offline-cached pools are future mitigations. Appendix E.1 gives the full air-gap justification and the standards catalogue.

MR designers can use offline batch SMS runs as a quantifiable design-feedback metric. We recommend a quarterly batch audit (about 0.5 person-day per quarter; detailed cost breakdown in Appendix E.2; estimates based on observed timings during this paper's 12-PUT campaign) rather than per-pull-request gating, because LLM API latency, cost non-determinism, and air-gap incompatibility together rule out the per-pull-request style. Appendix E.2 gives the quarterly-audit workflow with resource-cost table.

V&V documentation can carry SMS as research-grade supplementary evidence alongside code coverage and MR lists. We make no normative claim toward IEC, ISO, or ASME standards. An earlier draft proposed acceptance thresholds (aligned-cell SMS ≥ 0.20 or 0.30 plus C1_share ≤ 0.20); we removed them in revision because they have no normative backing and could be misread as enforcement-ready. Appendix E.3 records the conceptual complementarity with the code-verification scope of ASME V&V 20-2009 §3.

All three stakeholder classes consume the same single source of truth (`paper_numbers_v4.json` and `lrca_60cell_v4.json`) to avoid documentation fragmentation.

6 Threats to Validity

Table 20. Threats to Validity.

#	Threat	Class	Mitigation summary	Detail
R1	LLM generation reproducibility	Internal	Archived raw responses + frozen mutant pools (no user-facing seed; reproducibility by archival)	Appendix F.1
R2	E2 probabilistic approximation	Construct	K_eq = 1000 + Hoeffding bound; sweep deferred to future work	Appendix F.1
R3	Shared-infrastructure dependency	Internal	AVP version pinned to the upstream commit hash; embedded source	Appendix F.1
R4	LRCA multi-label boundary	Internal	Decision-tree priority + multi-label co-occurrence table	Appendix F.1
R5	12-PUT representativeness	External	shared infrastructure; Appendix B.1 coverage argument; scaling to industrial PUTs	Appendix F.2
R6	Cross-class statistical power	Conclusion	H3 framed exploratory; Friedman fallback; mixed-effects unavailable	Appendix F.2
R7	LLM homogeneity bias	Construct	3-LLM rotation + 20% manual sampling	Appendix F.2
R8	LLM-source distributional shift	External	DeepSeek 11/15 of overlaps; argument is categorical, not frequency	Appendix F.2

R9	Mutant-pool size	Internal	v3 mean 17.4, v4 mean 24.3 per PUT; effect intrinsic, not pool-dilution	Appendix F.1
R10	LLM non-determinism	Internal	De-dup, K = 10/20 repeats, raw-response store	Appendix F.1
R11	Reserved; retired pre-submission	Reserved	Numbering reserved to preserve cross-reference stability with prior P2 drafts	Reserved
R12	HOM equivalence	External	Confined claim to first-order syntactic tools; HOM testing deferred	Appendix F.2
R13	v3 vs v4 protocol asymmetry (dual-blind reviewer)	Internal	Quality not down (C1_share 0.164 → 0.209); follow-up to rerun dual-blind on v4	Appendix F.1
R14	Adjoint-arm reproduction fidelity and by-construction pools	Internal	Real defects reproduced through extracted diffs (no `arm64` wheel); arm is confirmatory on hand-constructed, low-cardinality pools, not a high-adequacy discovery; scipy #8900 is shared with the §4.8 boundary study	§4.9

R15	Real-defect evidence-slice boundary	External	Used only as a compact support slice for construct separation; reusable benchmark construction deferred outside this manuscript	§5.2

Final limitations. We list ten known limitations.

(a) **Equivalence determination is a probabilistic approximation.** Sampling K_eq = 1000 inputs is an engineering implementation of the undecidable equivalence problem, not a theorem-based decision. We give a Hoeffding-style upper bound on the false-equivalence probability in §2.3, but we did not execute a K_eq sweep over {500, 1000, 2000} for this submission. A follow-up study will run that sweep.

(b) **LLM generation homogeneity bias.** Even with three-LLM cross-source pooling, training-data overlap across Claude, GPT, and DeepSeek may produce similar blind spots. Three-LLM rotation and 20% manual sampling reduce but do not eliminate this risk. A stronger replication should add a fourth, independently trained LLM family; a self-hosted open-weight model is the natural candidate.

(c) **Limited statistical power for cross-class consistency.** The four-class sign test has df = 3, and the mixed-effects model returns Singular matrix at N = 60 over 12 PUTs. We treat H3 as exploratory and present RQ5 in four pieces: class means, sign test, Friedman, and a forest plot.

(d) **LRCA reports "likely" root causes.** The decision-tree priority C5 > C4 > C3 > C2 > C1 is an engineering choice. The reproducibility package preserves the multi-label co-occurrence table for every killed mutant as well as the priority-winning root cause; `root_cause` is best read as a likely cause rather than a definitive causal attribution.

(e) **The AVP is reused from the shared infrastructure.** Our reproducibility package embeds the AVP source code so that the package stays self-consistent under upstream evolution, but interface semantics may shift if the upstream infrastructure undergoes a major revision.

(f) **Epistemological scope versus engineering scope.** SMS measures semantic detection capability in the epistemological sense; engineering value is not evaluated here.

(g) **Signature simplification.** The `float` → `float` single-output signature is a substantive constraint, not a purely engineering trade-off, and it bounds the upper limit of mutant semantic complexity. A portability study should validate SMS on industrial-grade multi-output PUTs.

(h) **Air-gap incompatibility.** The mutant-generation workflow calls external LLM APIs and is incompatible with most regulated air-gapped V&V environments (IEC 60880, DO-178C, IEC 62304, ISO 26262). Pre-generated mutant pools can be reproduced offline, but generating new pools requires LLM access. Self-hosted open-weight LLMs and offline-cached pools are future mitigations.

(i) **No proof-carrying semantic-effect certificates.** The present workflow admits mutants through E1 ∧ E2, LRCA classification, and AST-normalised audit trails. These are empirical and structural certificates, not machine-checkable proof objects. In particular, we do not yet define an explicit abstraction function from concrete program semantics to scientific invariant domains, do not parameterise invariant families such as conservation, convergence, and stability as abstract-domain elements, and do not attach per-mutant proof, error expression, or proof-label certificates that bind a syntactic edit to a declared semantic-effect class. This is the main gap between SMS and a certificate-bearing semantic mutation framework.

(j) **Real-defect evidence is used narrowly.** The real-defect slice strengthens this paper's construct argument by showing that mutant kill-rate, semantic alignment, and real-defect detection can separate in practice. We do not claim benchmark construction, candidate curation, repository tooling, or corpus release as contributions of this paper; those are deferred to post-publication artifact work.

6.1 Statistical-modelling capacity at N = 60 cells / 12 PUTs

The mixed-effects primary model `sms ~ C(class) * C(operator) + (1 | put)` returned a Singular matrix at N = 60; the fallback `sms ~ C(class) + C(operator) + (1 | put)` had the PUT random-intercept variance hit boundary 0 (degenerate). Both failures are not numerical accidents; rather, they are evidence that 60 observations across 12 PUTs cannot identify an 11-dimensional fixed-effects structure plus 12-PUT random intercepts. This is a hard limit of the present design: **the experiment cannot, in principle, fit a hierarchical model with this fixed-effect dimensionality at this sample size**. Friedman tests serve as the non-parametric formal alternative, but they answer a structurally different question. Expansion to n_PUT $\geq$ 30 is the only path to a properly identified hierarchical model.

6.2 PUT-level zero-mass cohort

Figure 2 and §5.2 document that PUTs A1, B1, and D2 produce all-zero rows across the five MPs in the cross-source pool; that is, **for 25% of the PUTs, SMS gives zero signal at every operator-MP cell**. This is a substantive limitation of SMS as an adequacy metric on this PUT cohort: when the LRCA C1 mass collapses to zero across all aligned and cross slices for a PUT, SMS cannot distinguish strong from weak MR sets on that PUT. The phenomenon coexists with R_sem $\approx$ 1.0 (mutants are valid; §5.2), so the issue is not mutant generation but MR-design adequacy at the PUT level. MR-design refinement (per-PUT MP discovery) and the proposed pre-registered c-class primary-MP rule on a fresh dataset are the principal mitigation paths. We declare this as residual threat R14.

6.3 R13 protocol-asymmetry magnitude estimate

§3.9 declared protocol asymmetry R13: v3 used the Phase-1 dual-blind reviewer protocol (Claude generation + GPT-5.4 review + DeepSeek arbitration), while v4 passes V1-V4 mechanical gates only. To quantify the potential δ-shift attributable to this asymmetry without dual-blind, we offer a **rough order-of-magnitude estimate**: in the Phase-1 audit, dual-blind review filtered approximately 5-10% of LLM-generated mutants as semantically inconsistent; if these filtered mutants were systematically less effective at killing under MR_aligned (a plausible but unverified premise), the upper-bound δ-shift contribution from removing the reviewer step is bounded above by approximately **$\pm$0.03 to $\pm$0.05 on Cliff's δ** at n_aligned = 12. This bound is an order of magnitude larger than the observed -0.009 source-axis shift and therefore prevents reading the source-axis contrast as evidence of LLM-source-diversity inertness; a direct dual-blind v4 rerun (deferred to a follow-up study) would give a tighter quantification.

7 Conclusion

7.1 Findings summary

The empirical study supports the semantic-mutation construct in seven ways. The pre-registered thresholds are treated as stress tests of this instantiation, not as prerequisites for the SMS definition.

(a) SMS preserves the classical MS ratio while moving the operative definitions of mutant generation, equivalence, and killing into declared semantic strata. The 60-cell audit therefore evaluates how one concrete MR design observes semantic-effect fibers, rather than redefining mutation testing from scratch.

(b) The structural audit separates the semantic and syntactic mutant spaces. Across 12 PUTs, semantic mutants have 5.14% AST-normalised overlap with default cosmic-ray syntactic mutants; Hyperparameter, Structural Injection, and Trajectory Flip are 0/72, 0/33, and 0/54 under first-order defaults.

(c) Operator-MP alignment produces a positive but not large effect in this design. The H2 large-effect threshold is **not met under the pre-registered point-estimate criterion**: $\delta_{v3} = 0.323$, just below the Romano et al. [36] medium threshold of 0.330. The 49.1% stipulated power at the large-effect boundary shows why this verdict should be read as a point-estimate result, not as a proof that the true effect is below the large-effect threshold.

(d) Cross-source pooling changes mutant quality more than aligned-versus- cross separation. Holding the c-class primary metamorphic relation at the pre-registered MP5, three-LLM pooling under an identical prompt shifts Cliff's δ by only -0.009 (95% CI covers zero), while raising mean C1_share by 27%, class-c mean SMS by 91.5%, and class-d mean SMS by 38.3%.

(e) The MP structure is visible, but cross-class consistency is not yet stable. The Friedman main effect on MP differentiation is $\chi^2 = 15.30$, $p = 0.0041$, whereas the pre-registered H3 criteria are not met: the class-level sign test is 3/4 and CV(ΔSMS) exceeds 0.5. The Spearman correlation between SMS and Pattern Coverage is 0.163 at $n = 12$, so orthogonality remains a future hypothesis.

(f) The real-defect slice supports construct separation beyond the small PUT audit. Pattern-derived relations detect all 30 tabled real defects in the reported face, while aggregate mutant kill-rate gives only a modest advantage over literature-generic baselines and non-nesting cases remain. Thus kill-rate, semantic-stratum alignment, and real-defect detection should be interpreted as related but distinct constructs.

(g) The remaining failed thresholds delimit the current instantiation. Operator implementability H1 is not met because only Hyperparameter reaches ≥ 5 non-equivalent mutants on $\geq 9/12$ PUTs, and LRCA-mass H4 is not met because mean suspect_share is 0.791. These results identify where MR design and attribution need refinement; they do not change the degeneration theorem or the strong-boundary analysis.

7.2 Methodological contributions

This paper contributes a three-layer framework for domain-semantic mutation in single-output scientific computing kernels.

- **Layer 1 (§3.2).** Formal necessary conditions for semantic mutation, cross-function-boundary substitution, dependence on domain knowledge, change in algorithmic class, instantiated as five meta-operator classes: CE, OS, HP, TF, and SI (also written mut_C, mut_M, mut_G, mut_T, mut_F).
- **Layer 2 (§2.3).** The E1 ∧ E2 equivalence judgement, the conservative complete instantiation of the Layer-1 conditions, with an explicit trade-off against E1-alone and E2-alone variants.

- **Layer 3 (§3.5).** AST-normalised empirical traceability across all 12 PUTs: a 5.14% overall AST overlap with cosmic-ray defaults, and HP, SI, and TF unreachable at 0/0/0 under default first-order configurations (HOM not refuted; see §3.6(b)).

SMS is backward-compatible with the classical MS through §2.6 Theorem 9.1, which establishes almost-everywhere degeneration in the limit `L = L_equiv ∧ L_killed ∧ L_mut`. The 60-cell empirical audit reported in §4 is one demonstration following this backbone, not the paper's main contribution.

7.3 Future work

Seven directions follow from this study.

(a) A pre-registered c-class primary-MP rule on a fresh dataset.
(b) A differential-prompt LLM-diversity test using V_canonical, V_persona, and V_cot (full protocol in Appendix C.1.1).
(c) A scaling study at $n \geq 30$ PUTs for SMS-vs-Pattern-Coverage orthogonality and for HOM-equivalence empirics.
(d) An abstract-semantics certificate layer: define the abstraction α from concrete executions to scientific invariant domains, specify CE/OS/HP/TF/SI as semantic-effect predicates over those domains, and attach per-mutant proof objects, error expressions, or proof labels that certify class membership.
(e) A rerun of the dual-blind reviewer protocol on the full v4 grid, to separate protocol asymmetry (R13) from source diversity.
(f) Cross-language portability work (Python → JavaScript and Julia) building on Tip et al. [37].
(g) A self-hosted open-weight LLM pilot for air-gapped industrial deployment.

7.4 Data and code availability

All raw data, JSON SSOTs (paper_numbers_v3 / v4, rq2_cliffs_delta_v3 / v4_mp5, lrca_60cell, lrca_v4_mp5_recompute, rq2_power_stipulated, cosmic_ray_12put_ast_diff), mutant pools, AVP source, and analysis scripts are archived on Zenodo under DOI 10.5281/zenodo.20250664. An earlier version of this work is available on arXiv: arXiv:2605.17437 [cs.SE]. For peer review, an anonymized read-only mirror of the repository can be provided by the corresponding author upon Editor request, per journal guidelines. The repository structure follows `REPRODUCIBILITY.md` in the source tree. The cosmic-ray and mutmut operator versions referenced in §3.5, §3.6, and Appendix B.6 are pinned in `requirements-frozen.txt`.

Supplementary Material

Appendices A–I (notation and operator catalogue; experimental subjects and operator specialisations; experimental procedure details; statistical analysis details; deployment considerations; detailed threats-to-validity mitigation; the full SMS-to-MS degeneration proof; the adjoint extension arm; and a result-level real-defect evidence ledger) are provided as separate online supplementary material.

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CRediT authorship contribution statement

Meng Li: Conceptualization, Methodology, Software, Writing, original draft. **Xiaohua Yang**: Supervision, Formal analysis, Writing, review and editing. **Jie Liu**: Investigation, Validation. **Shiyu Yan**: Data curation, Visualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Generative AI Disclosure

The authors used AI tools under human direction for editorial proofreading, LaTeX maintenance, consistency checks, and review-simulation support. All research claims, methods, experimental data, analyses, and conclusions were authored, verified, and approved by the human authors.

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